

The Log-Quantile-Density Model of the Volatility Smile: Validity, Information, and Convention

Thibaut Delahaye

August 2026

Abstract

A volatility smile is a probability law expressed through the Black formula. This chapter constructs that law directly. The log-forward return is written as a monotone transport of a logistic variable, and the logarithm of the transport speed is expanded in Legendre polynomials of the percentile rank. Monotonicity, positive density, and unit mass are then automatic; a scalar shift imposes the martingale condition, and the sole remaining requirement is that the right endpoint speed be smaller than one. The endpoint speeds give the exact moment strip and the asymptotic Lee slopes. A single tail integral, the upper-share ledger, yields option prices, the density, the analytic calibration Jacobian, the variance-swap strike, and a complete test for calendar order. The construction separates what follows from probability, what is learned from quoted options, and what must be supplied by modeling convention. Theory, computation, and market examples are developed from the same representation.

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1 A smile is a probability law

A listed option market gives finitely many prices. A smile model must turn them into a complete curve: a price and an implied volatility at every strike, including strikes that were not quoted. The completed curve will be differentiated for digitals and densities, integrated for variance contracts, and compared with curves at other expiries. It is therefore not enough to draw a smooth line through the quotes.

The obstruction is elementary. The second strike derivative of a call price is a probability density. A curve that fits every quote can nevertheless make this derivative negative between two quotes. It then assigns a negative price to a butterfly and a negative mass to an interval of terminal prices. No numerical routine has malfunctioned; the curve was allowed to leave the space of probability laws.

The useful question is thus not “which curve fits well?” but “in which coordinates is every fitted curve already a law?” The answer developed here uses quantiles. In rank coordinates, unit mass is automatic and positive density reduces to monotonicity. Taking the logarithm of the quantile speed removes even that constraint. The resulting log-quantile-density (LQD) family provides unconstrained coordinates for a broad class of mean-one laws with exponential log-return tails.

1.1 Normalized market

Fix one expiry T . Let F be its forward and D its discount factor. Under the T -forward measure define

$$X = \log \frac{S_T}{F}, \quad Y = e^X = \frac{S_T}{F}, \quad \mathbb{E}[Y] = 1. \quad (1)$$

The last equality is the martingale condition. Write $k = \log(K/F)$ for log-moneyness. Normalized call and put prices are

$$c(k) = \mathbb{E}[(Y - e^k)^+], \quad P(k) = \mathbb{E}[(e^k - Y)^+], \quad c(k) - P(k) = 1 - e^k. \quad (2)$$

A dollar price is DF times its normalized price.

Let τ denote variance time: calendar time, possibly adjusted for the desk’s event-time convention. Total implied variance is $w(k) = \sigma(k)^2 \tau$. The normalized Black call is

$$B(k, w) = \Phi(d_+) - e^k \Phi(d_-), \quad d_{\pm} = -\frac{k}{\sqrt{w}} \pm \frac{\sqrt{w}}{2}, \quad (3)$$

and $w(k)$ is the root of $B(k, w(k)) = c(k)$ [2]. Rates and carry have now disappeared from the single-slice problem.

1.2 The static constraint set

Use normalized strike $y = e^k$. If Y has density f_Y , differentiation of the payoff gives

$$\frac{\partial c}{\partial y} = -\mathbb{P}(Y > y), \quad \frac{\partial^2 c}{\partial y^2} = f_Y(y) \geq 0. \quad (4)$$

This is the Breeden–Litzenberger identity [3]. It identifies the geometry of the call curve with the law of Y : decreasing slope is a digital probability and convexity is positive density.

The identity can be read using traded portfolios. For a strike spacing δ , the symmetric butterfly

$$c(y - \delta) - 2c(y) + c(y + \delta)$$

is the price of a long call below, a long call above, and two short calls at the center. Dividing by δ^2 and shrinking the spacing recovers $f_Y(y)$. A negative second difference is therefore not merely undesirable curvature; it is a negative price for a nonnegative terminal payoff. The probability statement and the static-arbitrage statement are the same.

Definition 1.1 (Arbitrage-free slice). An arbitrage-free slice is a call curve of the form (2) for a positive random variable Y with $\mathbb{E}[Y] = 1$. When Y has a density, this is equivalent to a decreasing convex function of strike y satisfying

$$(1 - y)^+ \leq c(y) \leq 1, \quad -1 \leq c'(y) \leq 0, \quad c(y) \rightarrow 0 \quad (y \rightarrow \infty),$$

with $c'(y) \rightarrow -1$ as $y \downarrow 0$ when there is no mass at zero.

Each boundary condition has a probabilistic source. The upper bound follows from $(Y - y)^+ \leq Y$ and $\mathbb{E}[Y] = 1$. The lower bound is Jensen's inequality, $(\mathbb{E}[Y] - y)^+ \leq \mathbb{E}[(Y - y)^+]$. The slope is a survival probability and must lie between -1 and zero. Finally, calls vanish at infinite strike because the integrable random variable Y carries no mean mass arbitrarily far in its tail. A valid call curve is thus exactly a compact encoding of a mean-one law.

Convexity belongs to y , not to k . Indeed,

$$\frac{\partial^2 c}{\partial k^2} = e^k \{e^k f_Y(e^k) - \mathbb{P}(Y > e^k)\},$$

which is negative far in the left wing for any ordinary density. A convexity test performed in log-strike will therefore reject correct call curves.

The same constraint becomes less transparent in volatility coordinates. Differentiating $c(k) = B(k, w(k))$ twice yields

$$f_X(k) = g_D(k) \frac{\varphi(d_-(k, w(k)))}{\sqrt{w(k)}}, \quad g_D(k) = \left(1 - \frac{kw'}{2w}\right)^2 - \frac{(w')^2}{4} \left(\frac{1}{w} + \frac{1}{4}\right) + \frac{w''}{2}. \quad (5)$$

Thus a volatility curve is butterfly-free precisely when $g_D(k) \geq 0$ for all k [8]. The condition is global, nonlinear, and involves two derivatives. Direct volatility models must enforce it; the representation below makes it an identity.

This explains why good residuals alone are weak evidence. Quotes constrain only finitely many values of w . The function g_D also depends on the slope and curvature between them, and may become negative in a narrow unquoted interval. Increasing the strike grid reduces but does not remove the logical problem: the optimizer still moves in a space containing invalid curves. The quantile construction reverses the order of work. It first chooses a law and only then asks which smile the Black formula assigns to it.

Two checks fix the normalization in (5). If the smile is flat, $w' = w'' = 0$, so $g_D \equiv 1$ and the remaining factor is exactly the Black density of X . At the money, the identities derived later in section 5 give $g_D(0) = \sqrt{w_0} f_X(0) / \varphi(\sqrt{w_0}/2)$. Thus $g_D \geq 0$ is not a penalty formula: it is positive density written in the volatility chart.

1.3 Notation ledger

The construction uses three horizontal axes. Strike y is the axis on which calls are convex; log-moneyness $k = \log y$ is the market axis; rank u and log-odds z are the model axes. For a rank $u \in (0, 1)$, the *odds* are $u/(1 - u)$ and the *log-odds* are

$$z = \log \frac{u}{1 - u}.$$

The function taking rank to log-odds is called the *logit*: $\text{logit}(u) = \log\{u/(1 - u)\}$. It maps $(0, 1)$ bijectively onto $\mathbb{R}$; its inverse is the logistic function $\Lambda(z) = 1/(1 + e^{-z})$. Thus u and z record the same percentile on two different scales. The recurring notation is collected below.

$X, Y = e^X$	log and gross forward return	$u, z = \text{logit } u$	rank and log-odds
$k, y = e^k$	log and normalized strike	Λ, ρ	logistic CDF and density
c, P	normalized call and put	$Q, q = Q'$	quantile and quantile density
$w = \sigma^2 \tau$	total implied variance	g	smooth log-speed in rank
x, m	transport and martingale shift	G	upper-share ledger
L, R, a_n	polynomial coordinates	λ_-, λ_+	endpoint speeds
u_k, z_k	strike rank and log-odds	u_*, f_*, Δ	ATM rank, density, digital gap
β_L, β_R	Lee wing slopes	$N, Z_{\max}$	basis order and grid half-width

Table 1: Notation used throughout the chapter. A prime denotes an ordinary derivative of a one-variable function; ∂ denotes a partial derivative. The time τ is always variance time.

1.4 What the chapter separates

Three questions recur, and they should not be confused. *Validity* asks whether a parameter vector defines a probability law. *Information* asks which features of that law are determined by the quoted strip. *Convention* selects the remainder: basis order, regularization, tail coordinates, and the scope of calendar control. The LQD coordinates settle the first question structurally. The later sections measure the second and state the third explicitly.

The development is progressive. Section 2 constructs the coordinates. Sections 3 and 4 establish their probabilistic content. Section 5 derives prices and local smile quantities from one integral. Section 6 turns the representation into a fitter, section 7 into a computation, and section 8 into a surface. Section 9 then shows what the construction says on market and synthetic data. Proofs and implementation details are collected in the appendices.

2 From ranks to LQD coordinates

2.1 The speed of a quantile

Let $Q(u)$ be the quantile function of X , with rank $u \in (0, 1)$. If U is uniform, then $Q(U)$ has the desired law. Hence probability one is built into the coordinate u , and the only remaining requirement is that Q increase.

Rank is a universal clock for distributions. The value $Q(0.01)$ is the first-percentile return whether the law is symmetric, skewed, or multimodal. Changing the law changes the returns assigned to the clock, not the clock itself. This is the main simplification: normalization of probability is handled once by the uniform variable rather than reimposed on every candidate density.

Write $q(u) = Q'(u)$ for the quantile density. Differentiating $F_X(Q(u)) = u$ gives

$$f_X(Q(u)) = \frac{1}{q(u)}. \quad (6)$$

The function q is the speed with which return space is traversed as rank increases. Slow motion packs probability into a narrow return interval; fast motion thins it. Since Q is increasing exactly when $q > 0$, an arbitrary real function may be used for $\log q$.

An unbounded distribution forces q to diverge at both endpoints. Here a *tail* means the probability lying beyond a remote threshold. We say that X has exponential tails, with positive scales λ_- and λ_+ , when

$$F_X(x) \sim C_- e^{x/\lambda_-} \quad (x \rightarrow -\infty), \quad 1 - F_X(x) \sim C_+ e^{-x/\lambda_+} \quad (x \rightarrow +\infty), \quad (7)$$

for constants $C_-, C_+ > 0$. Their logarithms therefore decay linearly in $|x|$. By contrast, a Gaussian tail has log-probability proportional to $-x^2$: it decays quadratically on the log-probability scale and hence faster than every exponential tail. For the exponential class, the

leading endpoint singularities of q are $1/u$ and $1/(1-u)$. Remove them explicitly:

$$\log q(u) = -\log u - \log(1-u) + g(u). \quad (8)$$

The remainder g is bounded for the exponential-tail class considered here.

The subtraction is not arbitrary. The left relation in (7) implies $Q(u) \sim \lambda_- \log u$ as $u \downarrow 0$ and $q(u) \sim \lambda_-/u$. The right endpoint is analogous. More precisely, if $X \sim N(0, s^2)$, then

$$F_X(x) \sim \frac{s}{|x|\sqrt{2\pi}} e^{-x^2/(2s^2)} \quad (x \rightarrow -\infty),$$

with the symmetric formula on the right. Its quantile density is $q(u) = s/\varphi(\Phi^{-1}(u))$ and, by the normal tail equivalent,

$$q(u) \sim \frac{s}{u\sqrt{2\log(1/u)}} \quad (u \downarrow 0).$$

Its leading $1/u$ singularity is removed, but the remainder still tends to $-\infty$. Exponential tails therefore lie inside the LQD chart, whereas Gaussian tails lie on a different boundary.

Now set

$$z = \text{logit } u = \log \frac{u}{1-u}, \quad u = \Lambda(z) = \frac{1}{1+e^{-z}}, \quad \rho(z) = \Lambda(z)\Lambda(-z). \quad (9)$$

The function Λ is the logistic CDF and $\rho = \Lambda'$ its density. Because $du/dz = u(1-u)$, the singularities in (8) cancel:

$$\frac{dQ}{dz} = q(u)u(1-u) = e^{g(u)}. \quad (10)$$

Thus the entire distribution is generated by integrating a positive, ordinary function on the log-odds line.

The coordinate also places every percentile at a finite location. The median is $z = 0$; the tenth and ninetieth percentiles are $z = \mp \log 9$; moving one unit in z multiplies the odds $u/(1-u)$ by e . Meanwhile $\rho(z) \sim e^{-|z|}$, so rank integrals become ordinary integrals on $\mathbb{R}$ with exponentially decaying weight. This is why the same coordinate is convenient both analytically and numerically.

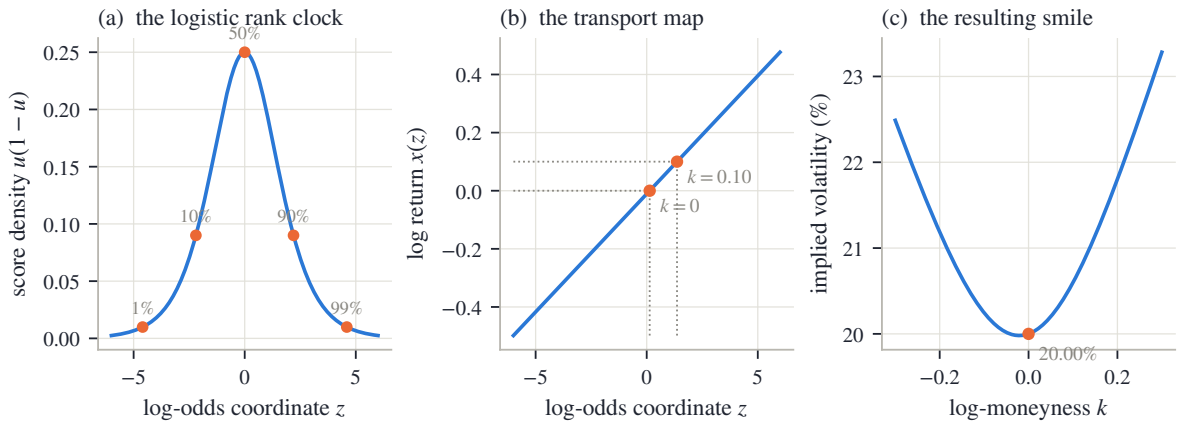


Figure 1: The transport construction for constant speed. Left: the logistic score and five percentile marks. Center: an affine map from log-odds to log-return, with two strike roots. Right: the implied-volatility smile priced from that law. The first two panels define the distribution; the last merely expresses it in market coordinates.

2.2 The LQD slice

Choose g to be a low-degree polynomial in rank. The shifted Legendre basis is convenient because it is orthogonal under the uniform rank measure. The constant and linear terms are written as an endpoint chord, leaving higher modes to describe the body.

Definition 2.1 (The LQD slice). Fix $N \geq 2$ and $\theta = (L, R, a_2, \dots, a_N) \in \mathbb{R}^{N+1}$. Let Z have CDF Λ and density ρ . The LQD slice is the law of

$$\boxed{X = x(Z), \quad x(z) = m + \int_0^z e^{g(\Lambda(t))} dt, \quad g(u) = (1-u)L + uR + \sum_{n=2}^N a_n P_n(1-2u),} \quad (11)$$

where P_n is the Legendre polynomial of degree n and m is chosen so that $\mathbb{E}[e^X] = 1$.

For every finite coefficient vector, $x'(z) = e^{g(\Lambda(z))} > 0$. Consequently

$$F_X(x(z)) = \Lambda(z), \quad f_X(x(z)) = \frac{\rho(z)}{x'(z)} = \Lambda(z)\Lambda(-z)e^{-g(\Lambda(z))} > 0. \quad (12)$$

The quantile is $Q(u) = x(\text{logit } u)$, and (10) recovers

$$q(u) = \frac{e^{g(u)}}{u(1-u)}. \quad (13)$$

Equations (11) and (13) are the transport and log-quantile-density descriptions of the same law.

Since a polynomial g is bounded on $[0, 1]$, its exponential is bounded above and away from zero. Therefore $x(z)$ tends to $-\infty$ and $+\infty$ with z and is a diffeomorphism of the line. The construction cannot fold, stop, or reverse. Positivity of the density in (12) is thus a geometric fact about the coordinate map, not a pointwise inequality checked after the fit.

The two endpoint values are

$$g(0) = L + \sum_{n \geq 2} a_n, \quad g(1) = R + \sum_{n \geq 2} (-1)^n a_n, \quad (14)$$

and their exponentials

$$\lambda_- = e^{g(0)}, \quad \lambda_+ = e^{g(1)} \quad (15)$$

are the endpoint speeds. If $\bar{x}(z) = \int_0^z e^{g(\Lambda(t))} dt$, then Lipschitz continuity of g and exponential convergence of Λ to its endpoints imply

$$\bar{x}(z) = b_- + \lambda_- z + O(e^z) \quad (z \rightarrow -\infty), \quad \bar{x}(z) = b_+ + \lambda_+ z + O(e^{-z}) \quad (z \rightarrow +\infty). \quad (16)$$

The endpoint speeds will therefore govern both tails.

The modes have a direct interpretation. Raising g stretches returns near the affected ranks and therefore lowers density there by (6). Even modes act symmetrically; odd modes transfer shape from one side to the other; higher modes work on finer rank scales. The raw coefficients are not pure body controls, since (14) couples each of them to one or both tails. The endpoint chart introduced in section 6 removes that coupling without changing the family.

Legendre polynomials are a coordinate choice, not part of the probability argument. Rank integrals use uniform weight on $[0, 1]$, under which Legendre modes are orthogonal. Their columns remain distinguishable in least-squares normal equations as order rises. Monomials span the same space but inherit Hilbert-matrix conditioning. Chebyshev modes would also be reasonable; the endpoint chord would remain useful because it names the two tail controls.

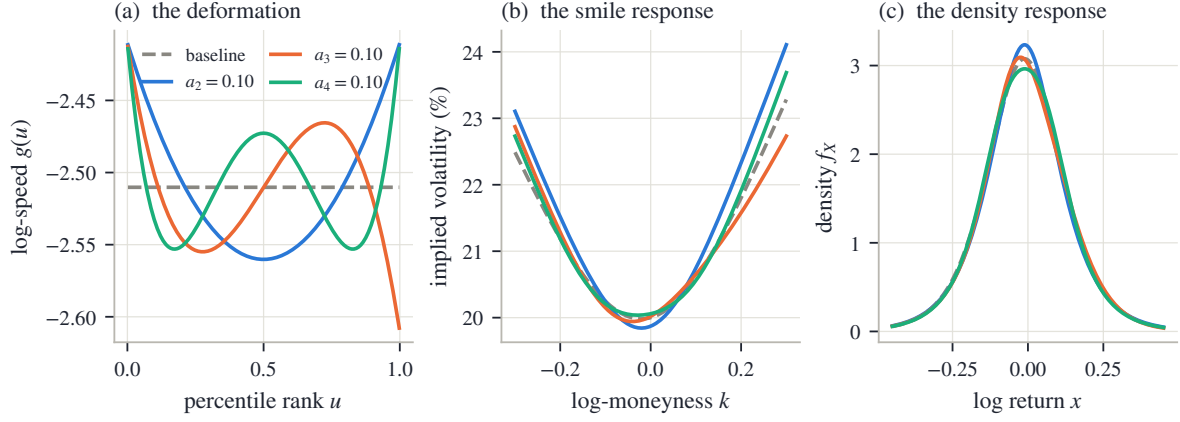


Figure 2: Three Legendre perturbations of the same reference law. Left: their action on log-speed g . Center: the response of implied volatility. Right: the response of the density. An even mode moves both endpoints in the same direction; an odd mode moves them in opposite directions. In all cases the density remains positive because the transport speed remains an exponential.

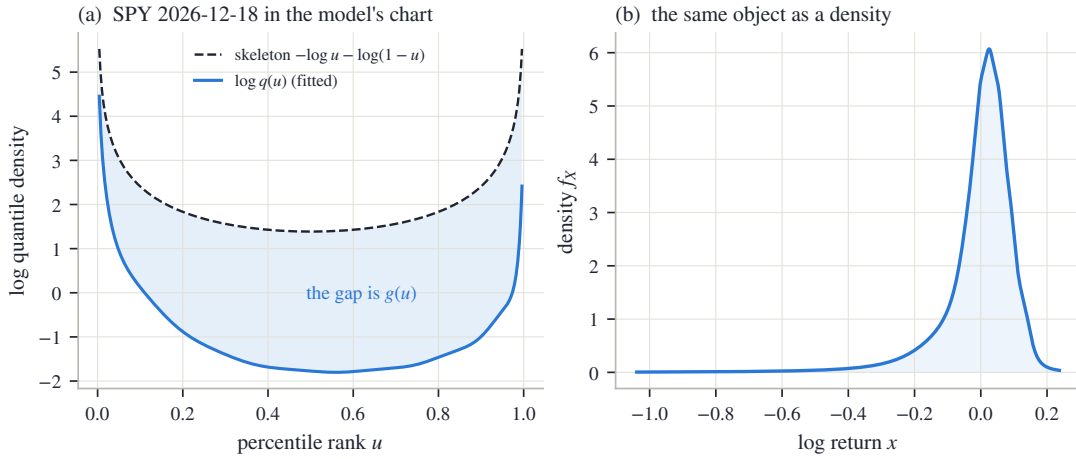


Figure 3: A fitted SPY December 2026 slice in the model's coordinates. Left: the universal boundary skeleton and the fitted $\log g$; their difference is g . Right: the density produced by the same transport. The quoted strikes occupy the central ranks; the continuation to the endpoints is extrapolation.

2.3 Martingale normalization and the finite-forward wall

Only a translation remains. Let

$$I = \int_{-\infty}^{\infty} e^{\bar{x}(z)} \rho(z) dz.$$

Proposition 2.2 (Martingale normalization). *If $I < \infty$, then $m = -\log I$ is the unique shift for which $\mathbb{E}[e^X] = 1$. Moreover, $I < \infty$ if and only if $\lambda_+ < 1$.*

Proof. Since $X = m + \bar{x}(Z)$, $\mathbb{E}[e^X] = e^m I$, giving the unique shift. At the right endpoint, (16) and $\rho(z) \sim e^{-z}$ make the integrand asymptotic to a positive constant times $e^{(\lambda_+-1)z}$; it is integrable exactly when $\lambda_+ < 1$. At the left endpoint it behaves as $e^{(1+\lambda_-)z}$, which is integrable for every $\lambda_- > 0$. $\square$

The set of admissible raw parameters is therefore

$$\Theta_N = \{\theta \in \mathbb{R}^{N+1} : \lambda_+ < 1\}. \quad (17)$$

It is open and has a single boundary. A logistic coordinate for λ_+ will make that boundary unreachable.

2.4 The exactly solvable slice

Set $L = R = \log s$, $a_n = 0$, with $0 < s < 1$. Then $x(z) = m + sz$ and X is logistic with scale s . Its moment-generating function is the beta integral

$$\mathbb{E}[e^{tZ}] = \int_0^1 \left(\frac{u}{1-u} \right)^t du = B(1+t, 1-t) = \Gamma(1+t)\Gamma(1-t) = \frac{\pi t}{\sin(\pi t)}, \quad |t| < 1. \quad (18)$$

Hence

$$m = -\log \frac{\pi s}{\sin(\pi s)}, \quad \text{Var}(X) = \frac{\pi^2 s^2}{3}.$$

Matching a target ATM total variance w_0 gives the cold start

$$s = \frac{\sqrt{3w_0}}{\pi}. \quad (19)$$

This seed matches variance, not ATM price. As $s \downarrow 0$, $m = -\pi^2 s^2/6 + O(s^4)$. Moreover,

$$\mathbb{E}[Z^+] = \int_0^\infty z \rho(z) dz = \sum_{n \geq 1} \frac{(-1)^{n-1}}{n} = \log 2,$$

where the logistic density is expanded as a geometric series on the positive half-line. Thus the LQD ATM call is $s \log 2 + O(s^2)$. Matching it to the small-variance Black price gives

$$\frac{w_{\text{implied}}}{w_0} \longrightarrow \frac{6(\log 2)^2}{\pi} \approx 0.918.$$

The seed is therefore about eight percent low in ATM variance, a known bias removed by calibration.

3 Validity and scope

The construction has two distinct claims. First, every admissible parameter vector defines an arbitrage-free slice. Second, increasing polynomial order fills out a specified class of laws. The first is exact at every order; the second determines the scope of the coordinates.

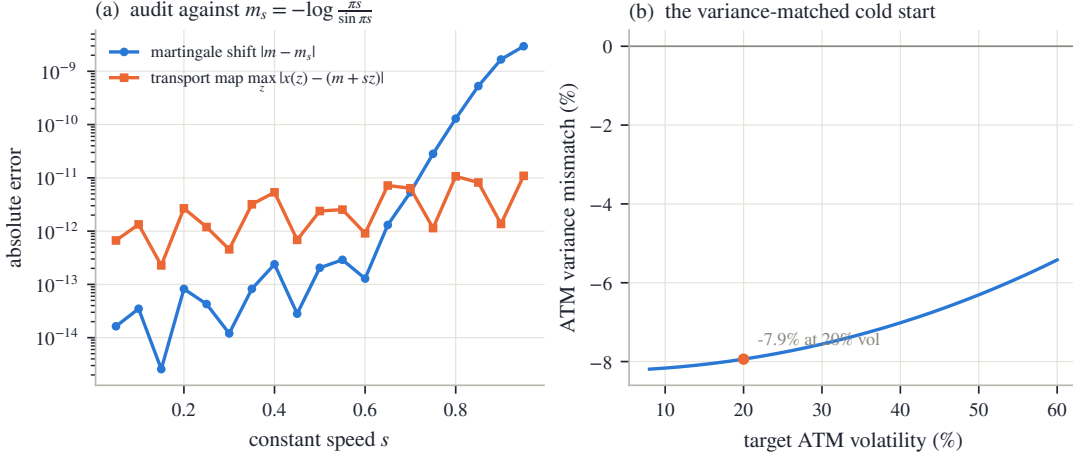


Figure 4: The solvable slice as an end-to-end numerical test. Left: errors in the computed transport and martingale shift against the closed forms from (18); the normalization becomes harder as s approaches the wall. Right: the variance-matched cold start and its derived small- s bias.

3.1 One-expiry validity

Proposition 3.1 (One-expiry no-arbitrage). *Let $\theta \in \Theta_N$ and price calls from the LQD slice (11). Then:*

- (i) X has a strictly positive continuous density on $\mathbb{R}$, and $Y = e^X$ has a strictly positive density on $(0, \infty)$ with $\mathbb{E}[Y] = 1$;
- (ii) as a function of strike y , the call is strictly decreasing and strictly convex, satisfies $(1 - y)^+ \leq c(y) \leq 1$, has $c'(y) = -\mathbb{P}(Y > y) \in (-1, 0)$, and tends to zero as $y \rightarrow \infty$;
- (iii) every nondegenerate finite-width butterfly has positive price, and put-call parity holds identically;
- (iv) as a function of log-strike k , the call is concave sufficiently far in the money.

Proof. Equation (12) gives a positive continuous density; the shift of proposition 2.2 gives unit mean. Rank integration gives

$$c(y) = \int_0^1 (e^{Q(u)} - y)^+ du.$$

Each integrand is decreasing and convex in y , and differentiation yields $c'(y) = -\mathbb{P}(Y > y)$. Positivity of the density makes both monotonicity and convexity strict. Jensen gives $(1 - y)^+ \leq c(y)$, while $c(y) \leq \mathbb{E}[Y] = 1$ and dominated convergence gives the right boundary. Butterfly positivity and parity follow immediately. Finally,

$$\frac{\partial^2 c}{\partial k^2} = e^k \{f_X(k) - \mathbb{P}(Y > e^k)\},$$

which is negative as $k \rightarrow -\infty$ because $f_X(k) \rightarrow 0$ and the probability tends to one. $\square$

The proposition is stronger than a dense-grid test. It covers every strike, including strikes between grid nodes and beyond the displayed smile, because the call is an expectation under a positive law. Calibration may move the coefficient vector arbitrarily inside Θ_N without approaching a hidden butterfly boundary. The numerical implementation must still preserve this continuous geometry; that separate question is treated by the certificates of section 7.2.

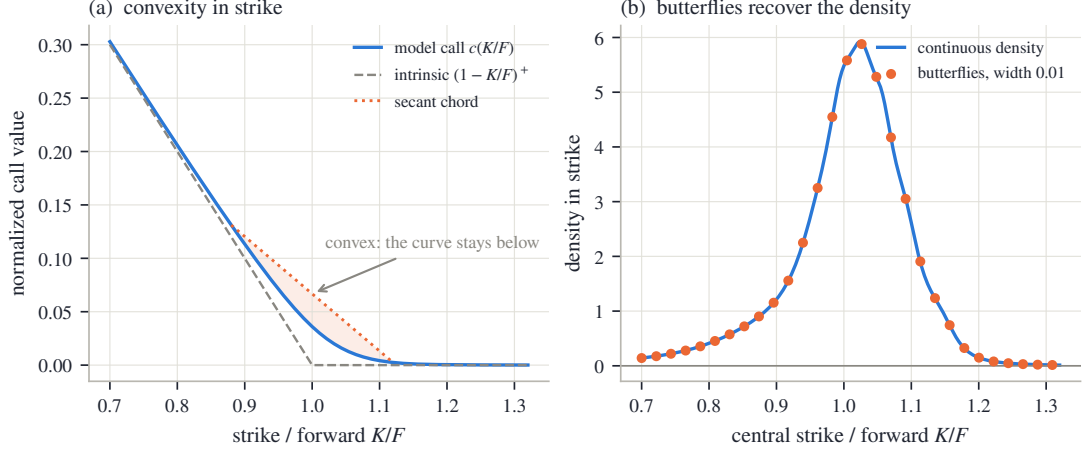


Figure 5: Validity on the fitted SPY December 2026 slice. Left: the call curve is convex in normalized strike and lies below its secant chord. Right: finite butterflies, divided by squared half-width, converge to the density. The geometry is produced by the law, not imposed during calibration.

3.2 Why the wall cannot be removed

Proposition 3.2 (Finite-forward wall). *Within the LQD family the martingale normalizer exists if and only if $\lambda_+ < 1$. More generally, if a law satisfies $\mathbb{P}(X > x) \geq \kappa e^{-x}$ for some $\kappa > 0$ and all sufficiently large x , then $\mathbb{E}[e^X] = \infty$.*

Proof. The first statement is proposition 2.2. For the second, at any large x_0 ,

$$\mathbb{E}[e^X] \geq \int_{e^{x_0}}^{\infty} \mathbb{P}(X > \log t) dt \geq \kappa \int_{e^{x_0}}^{\infty} \frac{dt}{t} = \infty.$$

□

The boundary is therefore probabilistic, not parametric. A right log-return tail with scale at least one cannot support a finite forward. There is no left counterpart because e^X is bounded on the negative half-line.

3.3 Coverage of the coordinates

The natural target class can be described without reference to polynomials. For a differentiable quantile Q^* define

$$g^*(u) = \log Q^{*'}(u) + \log u + \log(1 - u).$$

Continuity of g^* on $[0, 1]$ is exactly the condition that the law have a positive continuous density and exponential tails with finite nonzero endpoint scales.

Proposition 3.3 (Universality, with exclusions). *Let X^* be a mean-one law for which g^* extends continuously to $[0, 1]$, with $e^{g^*(1)} < 1$. Then:*

- (i) *There exist admissible LQD slices with $g_N \rightarrow g^*$ uniformly. Their quantiles converge uniformly on compact subsets of $(0, 1)$, their call curves converge uniformly over all strikes, and their gross-return laws converge in Wasserstein- p distance for every $1 \leq p < e^{-g^*(1)}$.*
- (ii) *No finite-order LQD slice has an atom, a zero-density interval, bounded support, mass at zero, or Gaussian-or-faster tails. These exclusions persist under uniformly convergent log-speeds. More precisely, let admissible g_N converge uniformly to g_∞ . If $g_\infty(1) < 0$, the limiting law remains in the class of part (i). If $g_\infty(1) = 0$, weak convergence can fail because mass escapes to $-\infty$; if a weak limit nevertheless exists, it has full support and a*

strictly positive continuous density, but sits at the finite-forward boundary. Its moments of order $r > 1$ diverge, and its mean may be smaller than one.

Here the one-dimensional Wasserstein metric has the concrete quantile form

$$W_p(\mu, \nu)^p = \int_0^1 |Q_\mu(u) - Q_\nu(u)|^p du.$$

The proof is in appendix A. Its approximation step is Weierstrass: polynomials approximate the continuous g^* uniformly. Since $g^*(1) < 0$, sufficiently accurate approximants remain inside the open admissible set. The tail margin supplies a common integrable envelope for normalizers, calls, and moments.

Uniform convergence of g is the correct strength for this result. It controls the transport speed at every rank, including both endpoints. Local approximation on the quoted ranks alone would say nothing about normalization or tail moments. Once a strict margin $g^*(1) < 0$ is fixed, the same exponential envelope controls all approximating normalizers and supplies uniform integrability of gross returns. This is what upgrades pointwise shape approximation to uniform convergence of call prices.

The exclusions follow from the same geometry. A positive continuous transport speed makes the quantile strictly increasing and unbounded. An atom would require a quantile jump; a gap, a flat interval; bounded support, a finite endpoint; and a Gaussian tail, $g(u) \rightarrow -\infty$ at an endpoint. None is available to a finite polynomial, nor to a uniform limit of bounded log-speeds.

Multimodality is not excluded. By (6), density modes are created where the transport slows and are separated where it accelerates. A positive speed can vary many times without ever becoming zero. Thus the family may represent a smooth double-hump density, but not a literal gap of zero density between the humps. Basis order determines how sharply the speed can vary, as the synthetic example in section 9 demonstrates.

Thus the LQD family is not a coordinate system for every probability law. It is a coordinate system for a flexible exponential-tail class. That class is large enough to approximate smooth unimodal and multimodal smile laws, while its boundary and exclusions remain explicit.

4 Tails, moments, and wings

The endpoint speeds introduced in (15) have an exact probabilistic meaning. This section follows the chain

$$\text{endpoint speed} \longrightarrow \text{tail scale} \longrightarrow \text{finite moments} \longrightarrow \text{asymptotic smile slope}.$$

4.1 Endpoint speeds are tail scales

Proposition 4.1 (Exponential tails). *For every admissible LQD slice there are constants $K_\pm > 0$ such that*

$$\mathbb{P}(X > x) \sim K_+ e^{-x/\lambda_+} \quad (x \rightarrow +\infty), \quad \mathbb{P}(X < x) \sim K_- e^{x/\lambda_-} \quad (x \rightarrow -\infty). \quad (20)$$

Equivalently, $\mathbb{P}(Y > y) \sim K_+ y^{-1/\lambda_+}$ and $\mathbb{P}(Y < y) \sim K_- y^{1/\lambda_-}$.

Proof. On the right, (16) gives $x(z) = m + b_+ + \lambda_+ z + O(e^{-z})$. Its inverse satisfies

$$z(x) = \frac{x - m - b_+}{\lambda_+} + O(e^{-z(x)}).$$

Since $1 - \Lambda(z) \sim e^{-z}$,

$$\mathbb{P}(X > x) = 1 - \Lambda(z(x)) \sim e^{(m+b_+)/\lambda_+} e^{-x/\lambda_+}.$$

The left tail is identical after reversing signs. □

The gross return therefore has power tails, while the log-return has exponential tails. The two endpoint coefficients control extrapolation in a directly readable unit: log-return distance per e -fold reduction in tail probability.

4.2 The moment strip

Every expectation may be written either in rank or log-odds:

$$\mathbb{E}[h(X)] = \int_0^1 h(Q(u)) du = \int_{-\infty}^{\infty} h(x(z)) \rho(z) dz. \quad (21)$$

Proposition 4.2 (Moment strip). *For an admissible LQD slice,*

$$\mathbb{E}[e^{rX}] < \infty \iff -\frac{1}{\lambda_-} < r < \frac{1}{\lambda_+}. \quad (22)$$

Consequently the last finite positive moment of Y beyond the first and the last finite inverse moment are

$$r_+^* = \frac{1}{\lambda_+} - 1, \quad r_-^* = \frac{1}{\lambda_-}.$$

Proof. In (21), the right-tail integrand for $h(x) = e^{rx}$ is asymptotic to $e^{(r\lambda_+ - 1)z}$; the left-tail integrand is asymptotic to $e^{(r\lambda_- + 1)z}$. Integrability gives exactly (22), with divergence at both boundary values. $\square$

The martingale condition is the point $r = 1$. It lies inside the strip exactly when $\lambda_+ < 1$, so normalization, the finite-forward wall, and the first moment are three forms of the same fact.

The asymmetry of the strip is financially important. A large λ_- may eliminate low-order inverse moments of Y without threatening the forward; negative log-returns contribute at most one to Y . A comparable right-tail scale would destroy the first positive moment and hence the forward itself. For the long NVDA example, $\lambda_- > 1$, so even $\mathbb{E}[Y^{-1}]$ diverges, while $\mathbb{E}[Y] = 1$ remains exact.

4.3 Closed Lee slopes

Lee's moment formula [12] relates the critical moments to the asymptotic slopes of total implied variance:

$$\beta_R = \limsup_{k \rightarrow +\infty} \frac{w(k)}{k} = \Psi(r_+^*), \quad \beta_L = \limsup_{k \rightarrow -\infty} \frac{w(k)}{|k|} = \Psi(r_-^*), \quad (23)$$

where

$$\Psi(r) = 2 - 4(\sqrt{r^2 + r} - r), \quad r \geq 0.$$

Proposition 4.3 (Speed-to-slope maps). *The LQD wing slopes are*

$$\beta_R = \frac{2\lambda_+}{(1 + \sqrt{1 - \lambda_+})^2}, \quad \beta_L = \frac{2\lambda_-}{(1 + \sqrt{1 + \lambda_-})^2}. \quad (24)$$

Both maps increase with their endpoint speed. For small speeds, $\beta_{\pm} \sim \lambda_{\pm}/2$; as $\lambda_+ \uparrow 1$, $\beta_R \uparrow 2$. For LQD slices the limsups in (23) are ordinary limits.

Proof. Substitute $r_+^* = 1/\lambda_+ - 1$ and $r_-^* = 1/\lambda_-$ into Ψ and rationalize. For example, with $s = \sqrt{1 - \lambda_+}$,

$$\Psi(1/\lambda_+ - 1) = 2 - \frac{4s}{1 + s} = \frac{2\lambda_+}{(1 + s)^2}.$$

The left formula is obtained with $t = \sqrt{1 + \lambda_-}$. Monotonicity and limits follow from the final expressions. By proposition 4.1, the return tails are regularly varying; the Benaim–Friz refinement of Lee's formula therefore gives convergence rather than only a limsup [1]. $\square$

For completeness, the left algebra is

$$\Psi(1/\lambda_-) = 2 - \frac{4(\sqrt{1+\lambda_-} - 1)}{\lambda_-} = \frac{2\lambda_-}{(1 + \sqrt{1+\lambda_-})^2}.$$

The formulas expose two useful scales. When λ is small, the wing slope is about half the endpoint speed. On the right, the map reaches Lee’s model-free ceiling 2 exactly when the first moment reaches its boundary. The tail coordinate and the no-arbitrage wing bound meet at the same point.

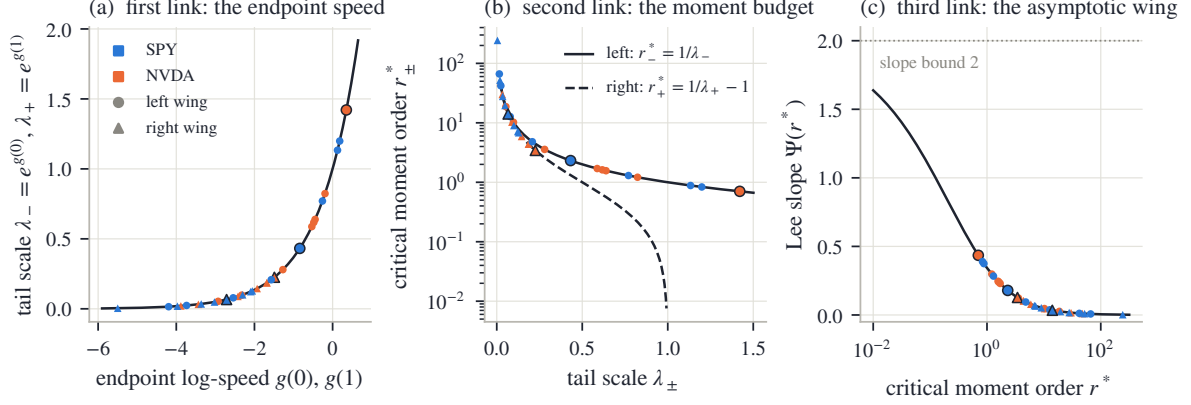


Figure 6: The tail chain. Left: endpoint speed determines the last finite moment; the right map reaches the finite-forward wall at $\lambda_+ = 1$. Center: Lee’s moment-to-slope function. Right: the closed compositions in (24). Markers show the fitted SPY December 2026 and NVDA December 2027 slices.

The marked market slices occupy only a small part of the admissible range. For SPY December 2026 the endpoint speeds are $\lambda_- = 0.432$ and $\lambda_+ = 0.066$, giving slopes $\beta_L = 0.179$ and $\beta_R = 0.034$. The long NVDA slice has $\lambda_- = 1.421$ and $\lambda_+ = 0.225$, with slopes 0.435 and 0.128. The heavier single-name tails appear as larger endpoint speeds before the smile is ever plotted.

4.4 The limit and the traded wing

The numbers $\beta_{\pm}$ describe limits, not quoted strikes. If $w(k) = \beta|k| + b + o(1)$, then $w(k)/|k| = \beta + b/|k| + o(1/|k|)$ and may approach its limit from either side. The sign of the intercept b is not fixed.

No finite option strip identifies an asymptotic decay rate by itself. In a fit, $\lambda_{\pm}$ are selected jointly by the outer quotes, basis order, regularization, and endpoint coordinates. The model makes this choice low-dimensional and interpretable, but it does not turn extrapolation into observation.

The finite-strike discrepancy is large enough to matter. On the SPY slice in fig. 7, the effective right slope is 1.69 times its limit at ten delta and 1.05 times at one delta. On the left, the corresponding ratios are 0.70 and 0.70, both below one. A limiting slope can therefore overstate one traded wing and understate the other on the same fitted law.

5 The ledger calculus

Once the transport x is known, a single tail integral contains nearly every quantity needed by the fitter.

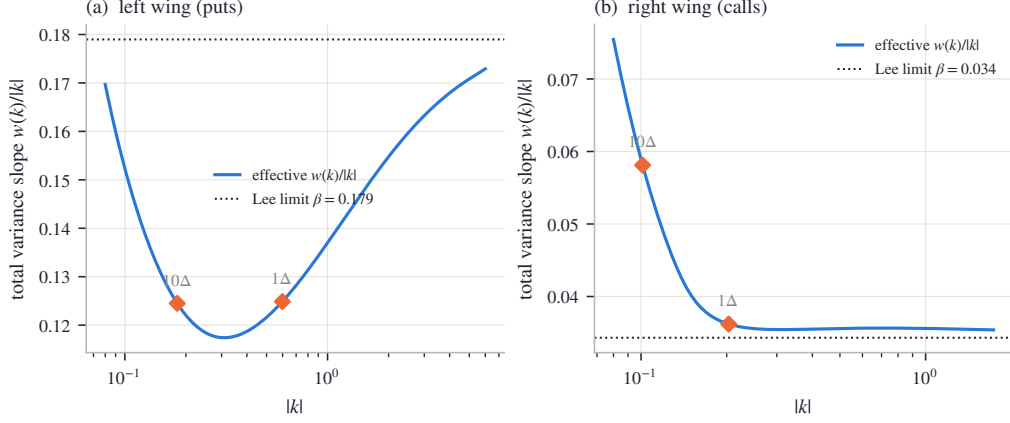


Figure 7: Effective variance slopes for the fitted SPY December 2026 slice. The left wing lies below its Lee limit at the marked deltas, while the right wing lies above. The asymptotic slopes are boundary conditions for extrapolation; they are not substitutes for prices at finite strikes.

5.1 The upper-share ledger

Definition 5.1 (Upper-share ledger). For an admissible slice define

$$G(z) = \int_z^\infty e^{x(t)} \rho(t) dt = \mathbb{E}[Y \mathbf{1}\{Z > z\}]. \quad (25)$$

The name is literal: $G(z)$ is the fraction of the unit forward carried by outcomes above log-odds level z . It satisfies

$$G(-\infty) = 1, \quad G(+\infty) = 0, \quad G'(z) = -e^{x(z)} \rho(z) < 0. \quad (26)$$

For strike k , let z_k be the unique root and u_k its rank:

$$x(z_k) = k, \quad u_k = \Lambda(z_k) = F_X(k). \quad (27)$$

Proposition 5.2 (Ledger pricing). For every admissible slice,

$$c(k) = G(z_k) - e^k(1 - u_k), \quad (28)$$

$$P(k) = e^k u_k - (1 - G(z_k)), \quad (29)$$

$$c'(k) = -e^k(1 - u_k), \quad (30)$$

$$c''(k) = -e^k(1 - u_k) + e^k \frac{\rho(z_k)}{x'(z_k)}, \quad (31)$$

$$f_X(k) = e^{-k} \{c''(k) - c'(k)\} = \frac{\rho(z_k)}{x'(z_k)}. \quad (32)$$

Proof. Because x increases, exercise is the event $Z > z_k$. Splitting asset and cash legs gives

$$c(k) = \mathbb{E}[Y \mathbf{1}\{Z > z_k\}] - e^k \mathbb{P}(Z > z_k) = G(z_k) - e^k(1 - u_k),$$

and the put follows on the complementary event. Differentiating the call, the moving-root term is

$$\{G'(z_k) + e^k \rho(z_k)\} \frac{dz_k}{dk} = 0$$

because $G'(z_k) = -e^{x(z_k)} \rho(z_k) = -e^k \rho(z_k)$. This yields (30). Since $dz_k/dk = 1/x'(z_k)$, a second derivative gives (31); subtracting (30) gives (32). $\square$

Pricing is thus the difference between the share delivered on exercise and the strike paid on the same event. The digital probability is $1 - u_k$, and the density is one division of two arrays already present in the transport. No price differentiation is required in the implementation.

The same rank change of variables prices any integrable terminal payoff:

$$\mathbb{E}[H(Y)] = \int_{-\infty}^{\infty} H(e^{x(z)})\rho(z) dz. \quad (33)$$

The ledger is the specialization in which $H(y) = y\mathbf{1}\{y > e^{x(z_0)}\}$ and the threshold z_0 is allowed to vary. Calls are unusually cheap because their payoff splits into this asset tail and an ordinary probability tail. Digitals retain only the probability tail; power and log contracts become moments of the transport. The whole payoff calculus therefore remains on one fixed rank measure.

The constant-speed slice gives a concrete first calculation. Choose its scale so that the six-month ATM volatility is exactly twenty percent. The solution is $s = 0.081250$ and the closed-form shift is $m = -0.010883$. For the call with $k = 0.10$, the strike rank is $u_k = 79.65\%$. The upper-share entry is $G(z_k) = 0.247221$, while the cash entry is $e^k(1 - u_k) = 0.224876$. Hence

$$c(0.10) = 0.247221 - 0.224876 = 0.022345,$$

corresponding to 20.60% implied volatility. The symmetric law produces a smile because its exponential return tails carry more remote-strike value than the lognormal matched at the money.

The fitted law also gives the log-contract variance directly:

$$w_{\text{vs}} = -2\mathbb{E}[\log Y] = -2 \int_{-\infty}^{\infty} x(z)\rho(z) dz. \quad (34)$$

This is the direct-law form of the familiar strip replication: the classical construction synthesizes $-\log Y$ from calls and puts over all strikes [4, 13], while here the law is already in hand and the same payoff is integrated against its rank measure. Under the usual continuous-path, continuous-monitoring identity this is the fair variance-swap strike; jumps and discrete sampling take their standard separate corrections.

5.2 The ATM microscope

At the money, the smile is determined locally by three quantities from the law. Let z_* solve $x(z_*) = 0$, and write

$$c_0 = G(z_*) - (1 - u_*), \quad u_* = \Lambda(z_*), \quad f_* = \frac{\rho(z_*)}{x'(z_*)}. \quad (35)$$

The ATM Black variance is

$$w_0 = 4 \left[\Phi^{-1} \left(\frac{1 + c_0}{2} \right) \right]^2, \quad d = \frac{\sqrt{w_0}}{2},$$

and define the digital mismatch

$$\Delta = u_* - \Phi(d). \quad (36)$$

Proposition 5.3 (Exact ATM handles). *The total-variance smile $B(k, w(k)) = c(k)$ satisfies*

$$w(0) = w_0, \quad w'(0) = \frac{2\sqrt{w_0}}{\varphi(d)}\Delta, \quad (37)$$

$$w''(0) = \frac{2\sqrt{w_0}}{\varphi(d)}f_* - 2 + \left(\frac{1}{2w_0} + \frac{1}{8} \right) w'(0)^2. \quad (38)$$

In volatility units, $\sigma(k) = \sqrt{w(k)/\tau}$,

$$\sigma(0) = \sqrt{w_0/\tau}, \quad \sigma'(0) = \frac{\Delta}{\varphi(d)\sqrt{\tau}}, \quad (39)$$

$$\sigma''(0) = \frac{1}{\sqrt{\tau}} \left[\frac{f_*}{\varphi(d)} - \frac{1}{\sqrt{w_0}} + \frac{\sqrt{w_0}}{4} \left(\frac{\Delta}{\varphi(d)} \right)^2 \right]. \quad (40)$$

The full differentiation is given in appendix A.2. The first derivative contains the main idea. At $(k, w) = (0, w_0)$,

$$\partial_k B = -(1 - \Phi(d)), \quad \partial_w B = \frac{\varphi(d)}{2\sqrt{w_0}}, \quad c'(0) = -(1 - u_*).$$

Implicit differentiation therefore gives

$$w'(0) = \frac{c'(0) - \partial_k B}{\partial_w B} = \frac{2\sqrt{w_0}}{\varphi(d)} \{u_* - \Phi(d)\}. \quad (41)$$

The model digital is $1 - u_*$; the matched flat-Black digital is $1 - \Phi(d)$. Hence Δ is Black digital minus model digital. A negative equity skew is exactly a model digital richer than the flat-Black digital. Level, digital, and density determine level, skew, and curvature.

Three sign checks make the interpretation concrete. For the matched Black law, $X \sim \mathcal{N}(-w_0/2, w_0)$, so $u_* = \mathbb{P}(X \leq 0) = \Phi(d)$ and the skew vanishes. If $\sigma'(0) < 0$, then $u_* < \Phi(d)$ and the model digital exceeds the Black digital. Conversely, an upward skew is exactly a cheaper model digital. The comparison is with $\Phi(d)$, not with one half: the martingale shift may place the forward rank on either side of the median even when the skew is small.

The curvature formula adds one new local fact: f_* . Holding ATM price and digital fixed, a larger density at the forward raises $w''(0)$. This matches the geometry of a peaked distribution: more mass close to the forward makes nearby options cheap relative to options a little farther away, and the smile bends upward more rapidly. The other terms in (38) are the Black-coordinate correction and a quadratic skew contribution. Price level, first CDF value, and first density value form a hierarchy for the three smile handles.

The identities also run in reverse. A target ATM level fixes c_0 ; a target skew fixes u_* through (37); and a target curvature fixes f_* through (38). These are local constraints on the law rather than finite differences of a plotted smile.

5.3 Envelope cancellation

Calibration requires derivatives with respect to every parameter. The call depends on a parameter both through G and through the moving root z_k . The same cancellation used for (30) removes the latter.

Theorem 5.4 (Envelope cancellation). *For every coordinate θ_j ,*

$$\frac{\partial c(k; \theta)}{\partial \theta_j} = \frac{\partial G(z; \theta)}{\partial \theta_j} \Big|_{z=z_k(\theta)}. \quad (42)$$

Proof. Differentiate (28):

$$\frac{\partial c}{\partial \theta_j} = \frac{\partial G}{\partial \theta_j} \Big|_{z_k} + \{G'(z_k) + e^k \rho(z_k)\} \frac{\partial z_k}{\partial \theta_j}.$$

The bracket vanishes by (26) and $x(z_k) = k$. □

This is an envelope theorem: the exercise boundary is the point where the payoff integrand vanishes, so its displacement has no first-order value.

Because g is affine in the coefficients, all columns are obtained by two cumulative passes. With $\bar{x} = x - m$ and $I = \int e^{\bar{x}} \rho$, let $b_j(u) = \partial g(u) / \partial \theta_j$. Then

$$\frac{\partial \bar{x}(z)}{\partial \theta_j} = \int_0^z e^{g(\Lambda(t))} b_j(\Lambda(t)) dt, \quad (43)$$

$$\frac{\partial m}{\partial \theta_j} = -\frac{1}{I} \int_{-\infty}^{\infty} e^{\bar{x}(z)} \frac{\partial \bar{x}(z)}{\partial \theta_j} \rho(z) dz, \quad (44)$$

$$\frac{\partial G(z)}{\partial \theta_j} = \int_z^{\infty} e^{x(t)} \left(\frac{\partial m}{\partial \theta_j} + \frac{\partial \bar{x}(t)}{\partial \theta_j} \right) \rho(t) dt. \quad (45)$$

The basis columns are fixed functions,

$$b_L(u) = 1 - u, \quad b_R(u) = u, \quad b_{a_n}(u) = P_n(1 - 2u).$$

The first pass differentiates the transport speed and integrates it. The normalization dot product removes the component that would change $\mathbb{E}[e^X]$. The reverse pass differentiates the upper-share tail. Every stage follows the direction of the corresponding primal build, so the derivative is obtained by array operations rather than a separate root-and-price routine for each coefficient.

The variance-swap row is obtained from the same sensitivity of x :

$$\frac{\partial w_{vs}}{\partial \theta_j} = -2 \int_{-\infty}^{\infty} \left(\frac{\partial m}{\partial \theta_j} + \frac{\partial \bar{x}(z)}{\partial \theta_j} \right) \rho(z) dz. \quad (46)$$

Both analytic and finite-difference Jacobians scale as $O(Pn_{\text{grid}})$ for $P = N + 1$ parameters, but the analytic route avoids $P + 1$ complete rebuilds. In the reference implementation it reaches the same optimum, with costs agreeing to 1.7×10^{-10} relative and parameters to 3.7×10^{-7} , in 1.41 – $1.76 \times$ less time. The worst relative disagreement with an independent central difference is 3.7×10^{-6} .

The theorem is exact for the continuous curves. Numerically, x and G are interpolated separately, so between nodes the price interpolant is not algebraically identical to the integral of the quantile interpolant. The analytic derivative inherits this fourth-order interpolation error. The finite-difference comparison in fig. 8 measures the discrete discrepancy instead of assuming the continuous cancellation transfers without loss.

It is worth counting the shared work. One build stores (x, x', G, G') on a fixed grid. A price is one root and two ledger entries; a density is one division; the log-contract variance is one weighted sum; the Jacobian reads ledger sensitivities at the same roots; and calendar order, in section 8, will compare two such arrays. These are not five approximations to the law; they are five queries of the same law.

6 Calibration

Validity has already been settled by the coordinates. Calibration may therefore concentrate on information: selecting, among admissible laws, one that represents the quoted strip without inventing unsupported detail.

6.1 A one-expiry objective

Suppose a slice contains quotes (k_i, σ_i) with $w_i = \sigma_i^2 \tau$. The reference fit minimizes

$$\sum_i \left(\frac{c(k_i; \theta) - B(k_i, w_i)}{\partial_\sigma B(k_i, w_i) + \eta} \right)^2 + \alpha \sum_{n \geq 4} n^{2\nu} a_n^2. \quad (47)$$

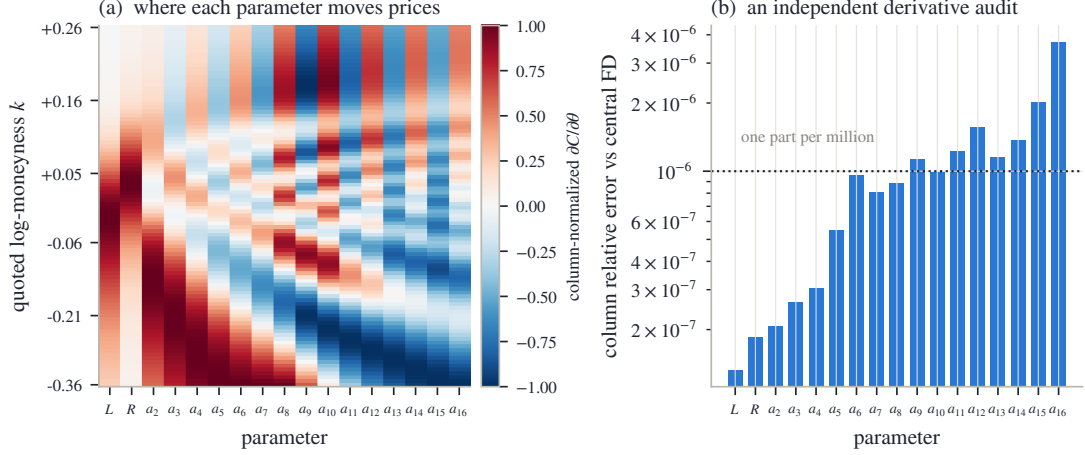


Figure 8: The analytic Jacobian on the fitted SPY December 2026 slice. Left: the normalized price-sensitivity pattern of every parameter across strikes. Right: columnwise disagreement with an independent central finite difference. The comparison tests the entire sensitivity pass by a separate route.

Each element of (47) answers one practical problem. The residual is computed in price, so implied-volatility inversion stays outside the optimization loop, where it cannot fail on an exploratory trial point. Division by the Black vega $\partial_\sigma B = \varphi(d_+) \sqrt{\tau}$ makes a central residual a volatility error — the unit in which quotes are judged — while the floor η lets the metric degrade to a scaled price error where vega vanishes, instead of assigning unbounded weight to a negligible option. The $n^{2\nu}$ ridge makes fine modes progressively more expensive but leaves a_2 and a_3 , the main curvature and asymmetry modes, unpenalized.

The reference objective also contains a smooth row $\log(1 + e^{50(\lambda_+ - 0.90)})$. The admissibility chart already prevents the wall from being reached; this row is instead an explicit prior against a nearly exhausted right-tail moment budget.

A quoted option is an interval, not necessarily a point. Three targets are useful. A *mid* fit uses (47) directly. A *band* fit uses the outside-band hinge together with a weak mid anchor. If $[p_i^{\text{lo}}, p_i^{\text{hi}}]$ is the price band and ϕ_i the vega normalizer, its two residuals are

$$\frac{(c_i - p_i^{\text{hi}})^+ + (p_i^{\text{lo}} - c_i)^+}{\phi_i}, \quad \sqrt{0.05} \frac{c_i - p_i^{\text{mid}}}{\phi_i}. \quad (48)$$

The squared hinge is continuously differentiable at a band edge, and the weak mid row — its weight deliberately small — selects one point when a wide band leaves a flat valley of equally admissible laws. A *haircut* fit first shrinks the volatility band by h :

$$\sigma_i^{\text{lo}} = \min(\sigma_i^{\text{bid}} + h, \sigma_i^{\text{mid}}), \quad \sigma_i^{\text{hi}} = \max(\sigma_i^{\text{mid}}, \sigma_i^{\text{ask}} - h). \quad (49)$$

If the original spread is narrower than $2h$, the band collapses to the mid instead of changing orientation. This occurs for every SPY quote in the frozen sample; the wider NVDA spreads retain live bands.

The two underliers illustrate both regimes. The SPY December node has median spread 4 vol bp, far inside the default $2h = 100$ vol-bp width, and none of its haircut bands survives. Its figures therefore show raw bid–ask intervals around a mid fit. On the featured NVDA nodes, 47% and 52% of the bands remain live. The same rule becomes a mid target on a liquid index and a genuine interval target on the wider single-name book.

The informational division can be read by location. Between liquid quotes a small change in the law moves observed prices and is strongly penalized: the strip determines the call curve there, and with it the ATM level, skew, curvature, and low-order moments. Near the edge of the

strip, data and regularization share control; beyond the outer quote, the endpoint speeds and the ridge supply most of the continuation. The curve itself is smooth — what changes gradually is the source of information, a transition made visible by plotting quote locations with the density as in fig. 3. Tail coordinates, ridge, and basis order should be reported with a fit because they make these choices.

6.2 Order and starting values

At order N the model has $N + 1$ parameters. The reference implementation uses the effective order

$$N_{\text{eff}} = \max \left(\min \left(N, \left\lfloor \frac{n_q}{2} \right\rfloor - 1 \right), \min(N, 6) \right), \quad (50)$$

where n_q is the quote count. The cap keeps the parameter count well below the number of observations; the floor leaves thin strips enough shape to form a usable smile. The rule is an identification convention, not a validity condition. Changing it changes inter-quote density detail while every resulting slice remains arbitrage-free.

Away from the floor, (50) requires roughly two quotes per body degree of freedom. This is not a theorem about optimal smoothing; it is a guard against inferring one oscillatory direction from each noisy observation. On the shortest NVDA slice, 17 quotes give $N_{\text{eff}} = 7$. The fit remains visibly less certain there rather than borrowing high-order detail from the basis.

Cold starts use the constant-speed value (19). In a maturity sweep, the previous expiry supplies a warm start. A multi-start check on the featured SPY and NVDA nodes used 10 randomized starts per node and found 1 basin, with worst fit-quality spread 0.00 vol bp. This supports the starting rule on those slices; event-dominated books may still require a broader search.

6.3 Coordinates for optimization

Three charts describe the same polynomial family.

1. The *raw chart* is $(L, R, a_2, \dots, a_N)$. It is simple, but (14) couples body modes to both tails.
2. The *endpoint chart* is $(\log \lambda_-, \log \lambda_+, a_2, \dots, a_N)$. Solving (14) for (L, R) is linear. Equivalently, each body basis function is replaced by an endpoint-vanishing combination. Body moves then hold both tail scales fixed.
3. The *logistic chart*, the reference default, writes $\lambda_+ = \text{expit } \xi$ with $\xi \in \mathbb{R}$. Every finite coordinate vector is admissible, and

$$\frac{d \log \lambda_+}{d \xi} = 1 - \lambda_+. \quad (51)$$

Steps shrink smoothly as the wall is approached.

The endpoint-vanishing body functions are explicit:

$$\tilde{P}_n(u) = P_n(1 - 2u) - (1 - u) - (-1)^n u, \quad n \geq 2. \quad (52)$$

They satisfy $\tilde{P}_n(0) = \tilde{P}_n(1) = 0$. A tail policy can therefore set (λ_-, λ_+) while calibration adjusts the body in a subspace that cannot move either endpoint.

The objective is identical in all three charts. Fits in the frozen sample agree across charts to 2.2×10^{-7} in the recovered raw parameters. In floating point, $\text{expit } \xi$ eventually rounds to one, so the implementation also enforces $\lambda_+ \leq 1 - 10^{-6}$.

6.4 Calibration sequence

The complete one-slice calculation is short enough to state as an algorithm.

1. Convert strikes and prices to (k_i, c_i) using the node's forward and discount factor. Map bid and ask volatilities through Black once to construct price bands and compute the quote vegas used for normalization.
2. Choose the effective order from (50) and select mid, band, or haircut residuals. This fixes the data part of the objective before optimization begins.
3. Construct a cold start from the ATM variance using (19), or map the previous expiry into the endpoint chart for a warm start. In either case, convert to logistic coordinates so all finite trial vectors are admissible.
4. At each trust-region step, build (x, x', G, G') on the fixed grid of section 7, price every quote by (28), and obtain the whole residual Jacobian from (43)–(45). Failed exponent-budget trials return a large residual rather than an undefined price.
5. After convergence, rebuild the accepted parameters on the dense grid. Evaluate ATM handles, tail moments, Lee slopes, the log-contract variance, and the numerical certificates (section 7.2) from this one final object.
6. If the slice belongs to a maturity sequence, add calendar rows against the accepted nearer slice and repeat. Report any residual common-support gap in normalized price units.

This order matters. Market conventions are fixed before the solve; probability validity holds during the solve; diagnostics are computed after the solve from the same law that produced the quoted prices. No separate post-processing smoother is inserted between calibration and publication.

7 Numerical realization

The theorems of the preceding sections concern the continuous objects x and G ; the fitter computes with discrete realizations of them. One grid build produces every array used above, analytic corrections handle what lies beyond the grid, and two certificates verify that discretization has preserved what the theory guarantees.

7.1 One grid build

All integrals use a symmetric log-odds grid $[-Z_{\max}, Z_{\max}]$, with $Z_{\max} = 40$. Optimization uses 2001 nodes; the accepted slice is rebuilt on 8001 nodes. A build performs four operations: evaluate g , integrate e^g cumulatively to form $\bar{x}$, compute the martingale shift, and integrate $e^x \rho$ backward to form G . Exact nodal derivatives $x' = e^g$ and $G' = -e^x \rho$ accompany the values. Cubic Hermite interpolation then gives fourth-order off-grid evaluation and a Newton-polished strike root.

The missing logistic tails are integrated analytically. To leading order,

$$\int_{Z_{\max}}^{\infty} e^{\bar{x}} \rho \, dz \simeq \frac{e^{\bar{x}(Z_{\max}) - Z_{\max}}}{1 - \lambda_+}, \quad \int_{-\infty}^{-Z_{\max}} e^{\bar{x}} \rho \, dz \simeq \frac{e^{\bar{x}(-Z_{\max}) - Z_{\max}}}{1 + \lambda_-}. \quad (53)$$

For a strike beyond the right return grid, continue the root along the tail asymptote,

$$z_R(k) = Z_{\max} + \frac{k - x(Z_{\max})}{\lambda_+}.$$

Both the ledger and cash leg are then single exponentials. Their difference is

$$c(k) \simeq e^{k-z_R(k)} \frac{\lambda_+}{1-\lambda_+}. \quad (54)$$

The left-wing continuation follows from parity.

To derive the factor in (54), restart the right-tail correction at $z_R(k)$, where $x = k$. The asset leg is approximately $e^{k-z_R(k)}/(1-\lambda_+)$ and the cash leg is $e^{k-z_R(k)}$. Subtraction leaves $\lambda_+/(1-\lambda_+)$. The continuation is therefore the tail form of the same asset-minus-cash identity (28), not an independent wing model.

Three floating-point rules are essential. First, compute $\rho(z) = \text{expit}(z) \text{expit}(-z)$ rather than $u(1-u)$, since u rounds to one for large positive z . Compute the cash leg as $\exp\{k - \log(1 + e^{z_k})\}$ using a stable log-sum-exp. Second, reject a trial if an interior value of g or $\bar{x}$ exceeds the exponent budget, even when the endpoints are admissible. Third, evaluate the Lee function at large r in the rationalized form

$$\Psi(r) = 2 - \frac{4}{\sqrt{1 + 1/r} + 1}, \quad (55)$$

which avoids subtracting nearly equal large numbers. The full build and sensitivity pseudocode appear in appendix B.

7.2 Certificates

The continuous theorems concern x and G ; numerical prices use their interpolants. Two inexpensive certificates connect the two levels. On each Hermite segment, the Fritsch–Carlson inequalities require both endpoint derivatives to lie between zero and three times the secant slope [6]. Applying them to x and $-G$ certifies monotone interpolation; segments below round-off are declared flat to a stated tolerance. Across the traded strike span, the analytic derivatives of $w(k)$ are also inserted in (5) on a dense grid and required to satisfy $g_D \geq -10^{-4}$.

An independent randomized battery prices 27 admissible slices across several orders and stress classes at sub-grid strikes. Its worst errors in call bounds, butterflies, digitals, and agreement with a four-times-finer grid are respectively 1.1×10^{-11} , 8.2×10^{-13} , 1.5×10^{-12} , and 2.3×10^{-9} in normalized price. The exact constant-speed test of fig. 4 independently checks transport, normalization, tail correction, and root evaluation. These numerical tests do not create validity; they verify that the discrete realization has preserved it to the reported tolerance.

The two-grid rebuild supplies a final resolution check. Parameters fitted on 2001 nodes and refitted on 8001 nodes agree to 1.4×10^{-6} . The accepted vector is rebuilt once on the dense grid, so prices, densities, ATM handles, and calendar ledgers all read the same final discrete object.

These diagnostics answer different questions and should not be collapsed into one flag. Quote residuals measure agreement with the strip; effective order and ridge describe the asserted inter-quote structure; endpoint speeds describe the extrapolation and its moment budget; monotonicity and belly margins certify that discretization preserved validity; calendar gaps describe compatibility across expiries. A fit can be excellent on one axis and weak on another.

8 Calendar order

Butterfly freedom is a property of one marginal law. A surface also requires later expiries to dominate earlier ones in convex order. Under deterministic rates and carry, this comparison has a simple form in normalized units.

Lemma 8.1 (Calendar ordering at equal log-moneyness). *Let $F_{0,t}$ be the deterministic forward curve and suppose $M_t = S_t/F_{0,t}$ is a positive martingale. For $T_1 < T_2$ and every k , the normalized calls satisfy*

$$c_1(k) \leq c_2(k). \quad (56)$$

Proof. Since $M_{T_i} = S_{T_i}/F_i = Y_i$, conditional Jensen gives

$$\begin{aligned} c_2(k) &= \mathbb{E}[(M_{T_2} - e^k)^+] \\ &\geq \mathbb{E}[(\mathbb{E}[M_{T_2} | \mathcal{F}_{T_1}] - e^k)^+] = \mathbb{E}[(M_{T_1} - e^k)^+] = c_1(k). \end{aligned}$$

□

Equal k means strikes $K_i = F_i e^k$, which are generally different dollar strikes. The comparison is a forward-scaled calendar relation, not the usual listed equal-strike spread. In probability language, the laws of Y_i have equal means and increase in convex order.

The scaling is essential. A violation at k is monetized by shorting $1/(D_1 F_1)$ calls at $K_1 = F_1 e^k$ and buying $1/(D_2 F_2)$ calls at $K_2 = F_2 e^k$. After normalization both payoffs are convex functions of the common martingale M_t . Without deterministic carry, one must first specify the numeraire and the objects being compared; equal log-moneyness alone is not a model-free calendar relation.

8.1 Calls and ledgers are conjugate

Lemma 8.2 (Ledger–call conjugacy). *For any admissible slice,*

$$\begin{aligned} c(k) &= \max_{z \in [-\infty, \infty]} \{G(z) - e^k(1 - \Lambda(z))\}, \\ G(z) &= \min_{k \in [-\infty, \infty]} \{c(k) + e^k(1 - \Lambda(z))\}. \end{aligned} \tag{57}$$

The optimizers are $z = z_k$ and $k = x(z)$, respectively.

Proof. For fixed k ,

$$G(z) - e^k(1 - \Lambda(z)) = \int_z^\infty (e^{x(t)} - e^k) \rho(t) dt.$$

The integrand is negative before z_k and positive after it. Starting the integral at z_k therefore maximizes it and gives the call price. The first identity implies $c(k) + e^k(1 - \Lambda(z)) \geq G(z)$ for every k , with equality at $k = x(z)$, proving the second. □

Geometrically, the first identity scans all possible exercise ranks. Before z_k , the integral includes outcomes where the payoff is negative; after z_k , it discards outcomes where the payoff is positive. The strike root is the unique point of tangency between call and ledger descriptions. The second identity reconstructs the ledger from the full call curve, so no information is lost in changing representation.

Theorem 8.3 (Calendar order is ledger order). *For two admissible mean-one slices,*

$$c_1(k) \leq c_2(k) \text{ for all } k \iff G_1(z) \leq G_2(z) \text{ for all } z. \tag{58}$$

Proof. If $G_1 \leq G_2$, the maximum representation in (57) gives $c_1 \leq c_2$. If $c_1 \leq c_2$, the minimum representation gives $G_1 \leq G_2$. Both implications are pointwise. □

Thus the continuum of calendar inequalities is equivalent to ordering two arrays already produced by the slice builds. This is also the classical convex-order criterion: by the results of Strassen and Kellerer, a family of equal-mean laws increasing in convex order can be realized as the marginals of a martingale [11, 16].

This is an existence statement. Ordered ledgers make the marginal laws compatible with some martingale dynamics; they neither select that dynamics nor determine forward smiles. The calendar problem treated here is exactly the static marginal problem.

8.2 A support-confined control

The theorem is an exact diagnostic. A finite calibration requires a chosen enforcement rule. When fitting a farther expiry after a nearer one, the reference objective adds

$$\sqrt{w_{\text{cal}}} \zeta_m \{c_{\text{near}}(k_m) - c_{\text{far}}(k_m)\}^+ \quad (59)$$

on a grid of strikes in the intersection of their quoted spans. The weights ζ_m taper to zero with a $\cos^2$ profile over the outer fifteen percent of that span. The number of nodes is $\max(49, 4N + 1)$.

The rows are imposed in price space, not ledger space. A ledger value at one rank integrates the entire upper tail, so constraining it couples the quoted region of one expiry to the unquoted continuation of another: the ledger is ideal for diagnosis precisely because it sees the whole law, and unsuitable as a constraint for the same reason. In a reproduced stress comparison, a full-grid ledger floor moved the worst quote error from 10.6 to 1095.0 vol bp; fixed-strike call rows, each a specified payoff in the common quoted region, localize the control. The weight is finite, and any remaining violation is reported as a price gap and as a fraction of the local bid–ask width. Calendar order is thus controlled on common quoted support; outside it, the ledger comparison remains a diagnostic of the extrapolations.

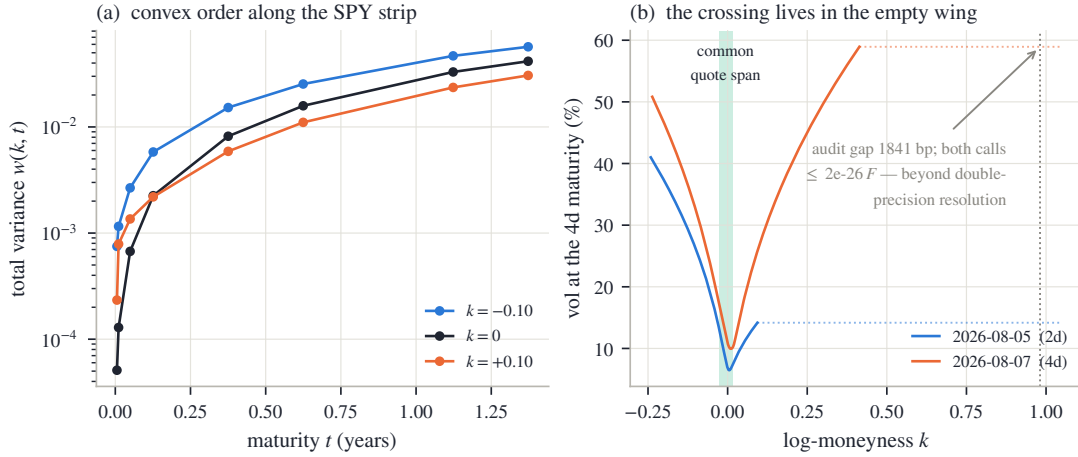


Figure 9: Calendar order in the frozen SPY sample. Left: total variance is ordered through maturity at representative log-moneyness values. Right: the two shortest expiries are ordered on their common quoted span, while a volatility-space crossing appears only in a numerically empty far wing. The price-space gap there is at round-off scale.

For the two shortest SPY expiries, a full-grid volatility audit flags a 1840.6 vol-bp crossing at $k = +0.981$, far outside the common quote span $[-0.028, +0.016]$. The corresponding calls are 5.4×10^{-107} and 1.6×10^{-26} of forward. A direct price audit finds a worst violation of only 2.2×10^{-15} . The crossing is a disagreement between extrapolations in a region where Black inversion is ill-conditioned, not an economically available calendar trade.

9 Examples and the model contract

All market examples use one frozen after-hours snapshot of SPY and NVDA listed options from the 2026-08-03 US session. Each underlier has eight expiries, for 16 fitted nodes in total. The common recipe is order $N = 16$ subject to (50), the logistic chart, the haircut target (49), ridge $\alpha = 10^{-6}$, and the calendar rows (59). No node is tuned separately. Across the sample the median quote error is 5.9 vol bp and the worst is 24.6 vol bp.

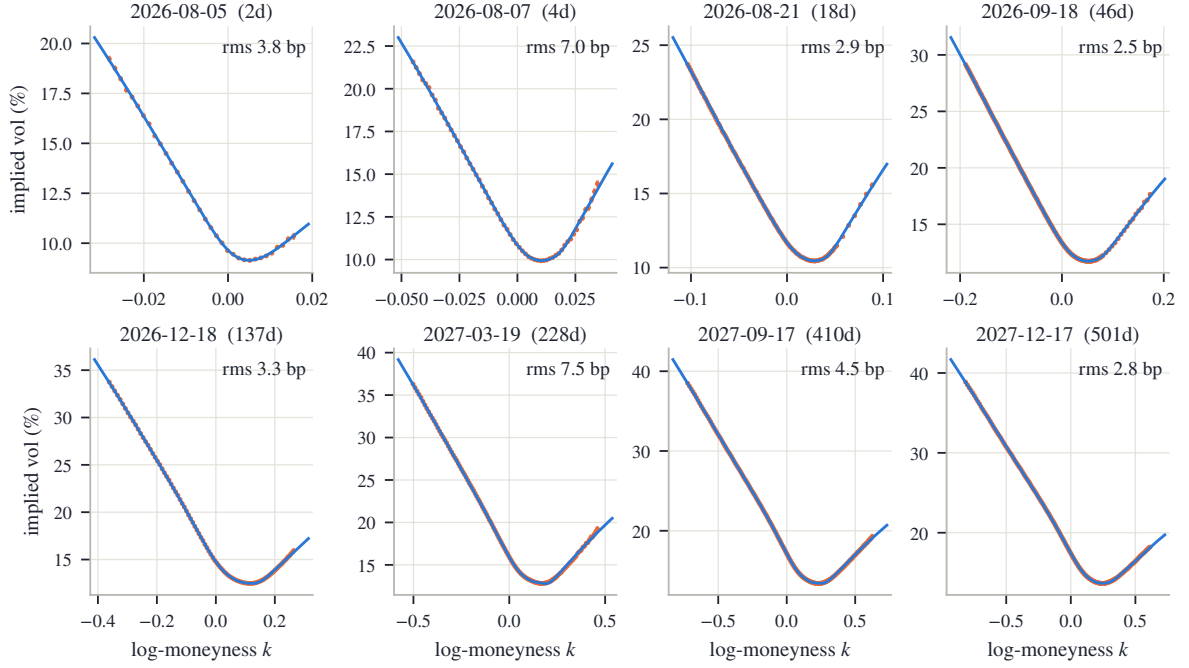


Figure 10: Eight SPY expiries fitted with one recipe. Each panel shows raw bid-ask quotes, mids, the fitted LQD slice, and rms error. The short end is steeper and more localized; the long end broadens smoothly. Beyond the last quote, each curve is determined by the fitted exponential-tail convention.

9.1 SPY: a term structure and one slice

The gallery separates recurring market shape from the completion chosen by the model. Through the quoted region the slices follow the term structure of level and skew; outside it the two endpoint speeds continue the curve. The boundary is most visible at the short end, where quoted spans are narrow in log-moneyness: the sharp local skew is supported by quotes, while the remote upward turn is the exponential-tail completion. At longer maturities the span widens and a larger fraction of the displayed smile is directly informed by prices. The median error across the eight slices is 3.6 vol bp; the worst is 7.5 vol bp.

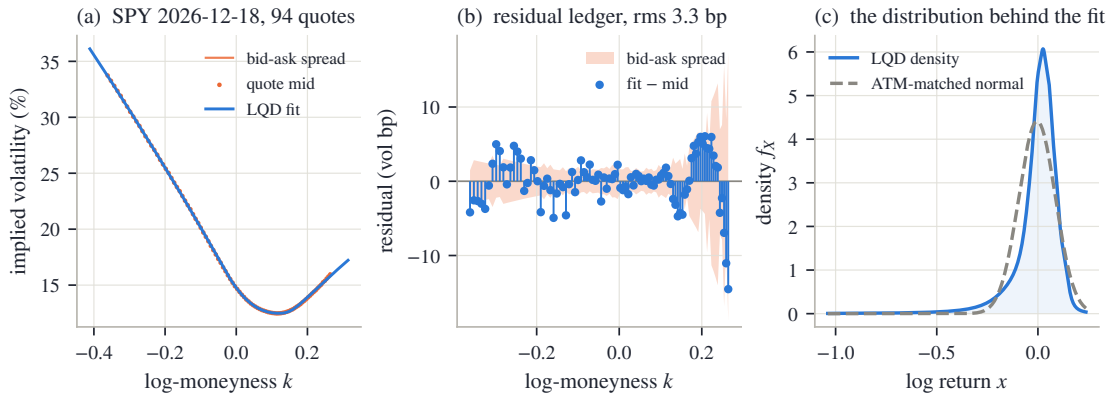


Figure 11: The SPY December 2026 slice. Left: quotes and fitted volatility. Center: signed residuals in vol bp. Right: the fitted density against an ATM-matched normal. The density view makes the negative skew visible as a transfer of mass toward the left tail.

The December 2026 slice contains 94 quotes and fits them to 3.3 vol bp rms. Its exact ATM quantities from proposition 5.3 are volatility 14.76%, skew -0.430 , and curvature 4.40. The

forward rank is 41.31%%, below the matched Black rank; hence $\Delta < 0$, in agreement with the negative skew. The log-contract variance corresponds to 19.84% annualized volatility.

The three panels answer different questions. The first tests the nonlinear map from law to implied volatility. The second asks whether residuals retain systematic strike structure. The third removes the Black coordinate and shows the fitted object itself: a left-heavy density. The ATM-matched normal has the same central price level but not the same digital or tail allocation, and therefore cannot reproduce the skew.

The ledger also makes an individual option transparent. For the listed December 800 call, $k = 0.0409$ and the strike rank is 65.27%%. The upper-share and cash entries are 0.378366 and 0.361834, so

$$c(0.0409) = 0.378366 - 0.361834 = 0.016532.$$

This is \$12.70 per share and corresponds to 13.37% implied volatility.

9.2 NVDA and the order guard

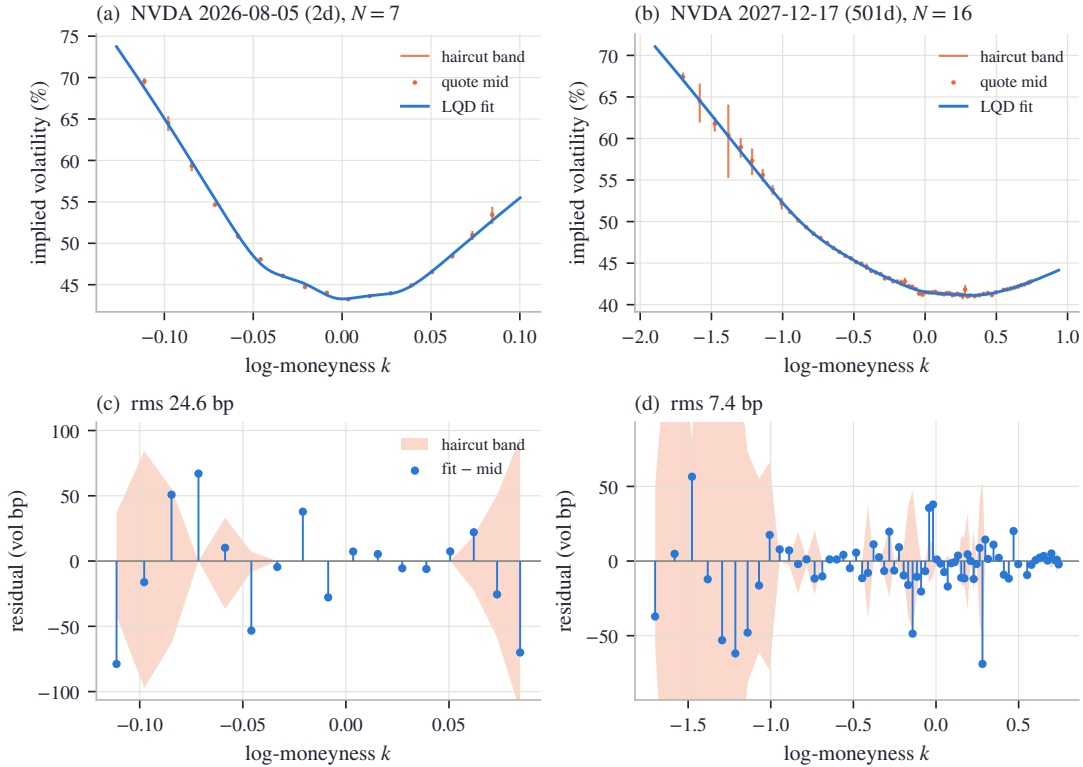


Figure 12: Two NVDA slices. Left: the shortest expiry, with 17 quotes and effective order $N_{\text{eff}} = 7$. Right: the December 2027 expiry at full order. Wider NVDA spreads leave live haircut bands and permit visibly larger residual variation than SPY.

The shortest node fits at 24.6 vol bp rms; the long node, with 69 quotes, at 7.4 vol bp. The latter has heavier fitted tails than SPY. In particular its left endpoint speed exceeds one, which is admissible: only the right endpoint is constrained by the finite-forward wall. The consequence is a last finite inverse-moment order 0.70 below one.

The shortest node is also the clearest example of the order guard. A thin strip cannot distinguish sixteen smooth modes, even though an optimizer can assign values to them. Reducing N_{eff} converts that lack of information into visible residual uncertainty instead of invisible oscillation between quotes. The long node has enough observations to use the full order and separate body curvature from tail continuation.

9.3 What basis order asserts

The synthetic benchmark is a martingale-normalized mixture of two lognormal components. It has a genuinely bimodal density but only gentle shoulder structure in implied volatility.

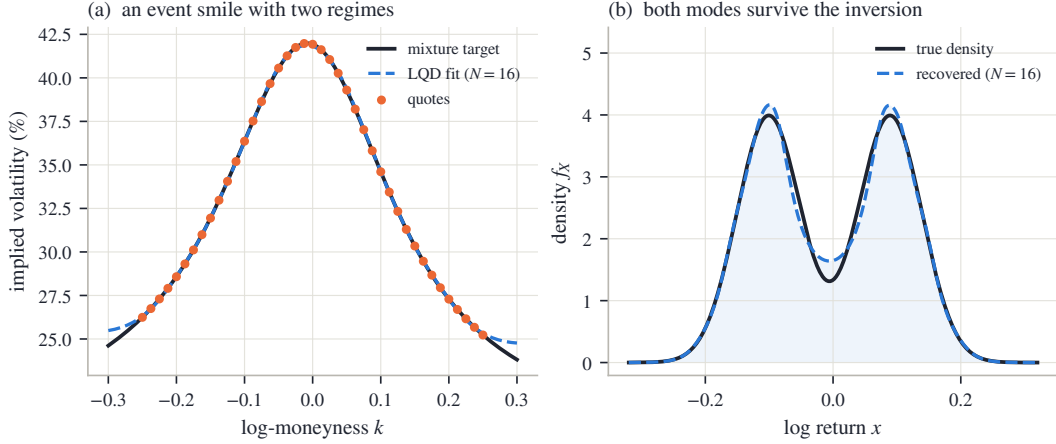


Figure 13: The double-hump benchmark at $N = 16$. Left: the fitted smile matches the synthetic quotes to 4.8 vol bp rms. Right: the fitted density recovers both modes, the trough, and the asymmetry of the target.

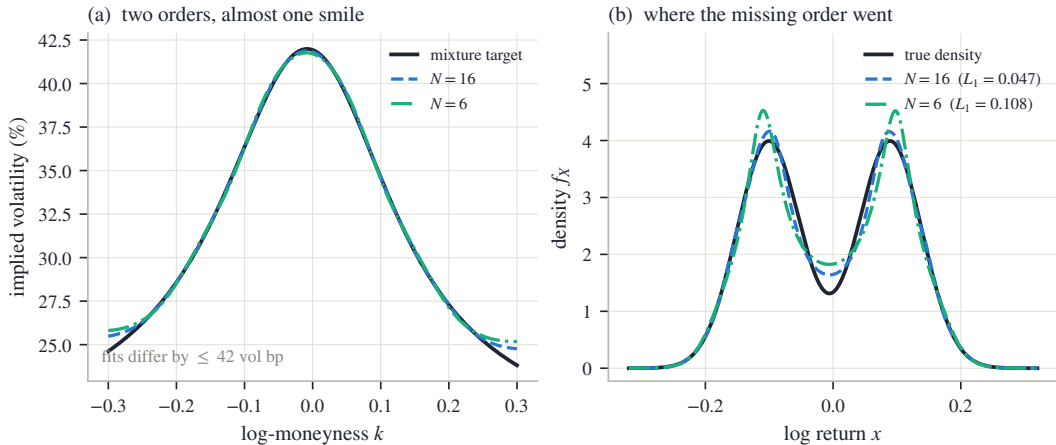


Figure 14: The same target at orders 6 and 16. Left: the implied-volatility curves nearly coincide over the quoted region. Right: the densities do not; the lower-order fit displaces the modes and fills part of the trough. Basis order controls how much inter-quote structure the model is allowed to assert.

At $N = 16$ the fitted mode locations are -0.099 and 0.088, against true locations -0.101 and 0.089. At $N = 6$ both modes remain, but their shape is distorted. The L^1 density error rises from 0.047 to 0.108, even though the two implied-volatility curves differ by at most 41.8 vol bp on the comparison grid. Option quotes can be nearly indifferent between laws that price density-sensitive structures differently.

The lesson is not that high order is always preferable. At $N = 16$ the synthetic target is known to contain fine structure, so extra modes recover a feature known to be real. In market data the truth is not supplied. High order then grants the model permission to assert finer inter-quote density shape. Digitals and narrow butterflies are sensitive to that permission; broad vanilla prices may not be. Basis order should be chosen for the downstream instruments, not solely for the in-sample volatility residual.

9.4 Position among existing models

The construction joins several classical ideas. Breeden–Litzenberger identify option prices with a density [3]; Lee and Benaim–Friz connect moments with wing slopes [1, 12]; quantile modeling supplies the rank coordinate [9, 14]; and Petersen–Müller use the same log-quantile-density transform to place density-valued data in an unconstrained function space [15]. The metalog family is a related direct quantile expansion [10].

SVI and SSVI work in volatility coordinates and characterize admissible parameter regions [7, 8]; arbitrage-free smoothing works directly with constrained prices [5]. LQD instead makes the probability law primary. Its contribution is the combined architecture: structural one-slice validity, readable tail coordinates, and one ledger for pricing, differentiation, density, variance, and calendar order.

9.5 The contract

Two things remain: how to read a fitted slice, and what precisely is claimed.

A fitted slice should be read in the same order as the theory. First inspect quote residuals over the retained span. Then read the three ATM handles as local level, digital, and density information. Next inspect $g(u)$ and the density over the ranks occupied by quotes. Only after that should one read $\lambda_{\pm}$, the moment strip, and Lee slopes, since these describe the extrapolated endpoints. Finally compare the ledger with neighboring expiries. This sequence prevents a tail convention or a calendar diagnostic from being mistaken for a directly observed quote.

Structural	Numerically controlled	Outside the claim
Positive density, unit mean, and butterfly-free calls for every admissible slice	Interpolation monotonicity, traded-range density, and randomized price audits	Atoms, mass at zero, density gaps, bounded support, and Gaussian-or-faster tails
Exact exponential tail scales, moment strip, and Lee limits	Tail quadrature, exponent budgets, grid convergence, and finite-strike wing prices	Tail parameters as observations, or asymptotic slopes as finite-strike quotes
Exact ATM handles, analytic Jacobian, and ledger characterization of calendar order	Quote fit, basis-order control, multi-start checks, and calendar order on common quoted support	Dynamics, jump decomposition, realized-variance corrections, and calendar structure outside the controlled support

Table 2: The model contract: theorem, numerical control, and excluded scope.

The central distinction is now visible in one place. Probability supplies validity. Quotes supply information over a finite strip. Basis, ridge, tails, and calendar scope supply convention elsewhere. The value of the LQD representation is not that convention disappears, but that it can no longer be mistaken for a theorem or for market information.

A Proofs deferred from the main text

A.1 Derivation of the volatility-space density factor

Let $c(k) = B(k, w(k))$. For general (k, w) , the identity $e^k \varphi(d_-) = \varphi(d_+)$ gives

$$\partial_k B = -e^k \Phi(d_-), \quad \partial_w B = \frac{\varphi(d_+)}{2\sqrt{w}}.$$

The second partials required by the chain rule are

$$\partial_{kk} B = -e^k \Phi(d_-) + \frac{\varphi(d_+)}{\sqrt{w}},$$

$$\begin{aligned}\partial_{kw}B &= \frac{d_+\varphi(d_+)}{2w}, \\ \partial_{ww}B &= -\frac{d_+\varphi(d_+)}{2\sqrt{w}}\partial_w d_+ - \frac{\varphi(d_+)}{4w^{3/2}}, \quad \partial_w d_+ = \frac{k}{2w^{3/2}} + \frac{1}{4\sqrt{w}}.\end{aligned}$$

Differentiate $c(k)$ twice and subtract the first derivative:

$$c'' - c' = (\partial_{kk}B - \partial_k B) + (2\partial_{kw}B - \partial_w B)w' + \partial_{ww}B(w')^2 + \partial_w B w''.$$

Factor $\varphi(d_+)/\sqrt{w}$. Since $d_+ = -k/\sqrt{w} + \sqrt{w}/2$,

$$2\partial_{kw}B - \partial_w B = -\frac{k\varphi(d_+)}{w^{3/2}},$$

and a direct multiplication gives

$$d_+\partial_w d_+ = -\frac{k^2}{2w^2} + \frac{1}{8}.$$

Therefore

$$\begin{aligned}c'' - c' &= \frac{\varphi(d_+)}{\sqrt{w}} \left[1 - \frac{kw'}{w} + \left(\frac{k^2}{4w^2} - \frac{1}{4w} - \frac{1}{16} \right) (w')^2 + \frac{w''}{2} \right] \\ &= \frac{\varphi(d_+)}{\sqrt{w}} \left[\left(1 - \frac{kw'}{2w} \right)^2 - \frac{(w')^2}{4} \left(\frac{1}{w} + \frac{1}{4} \right) + \frac{w''}{2} \right].\end{aligned}$$

Finally, $f_X(k) = e^{-k}(c'' - c')$ and $e^{-k}\varphi(d_+) = \varphi(d_-)$, which proves (5).

A.2 Proof of the ATM formulas

The smile is defined by $B(k, w(k)) = c(k)$. Put $w_0 = w(0)$ and $d = \sqrt{w_0}/2$. The identity $e^k\varphi(d_-) = \varphi(d_+)$ follows from $d_+^2 - d_-^2 = -2k$ and gives

$$\partial_k B = -e^k\Phi(d_-), \quad \partial_w B = \frac{\varphi(d_+)}{2\sqrt{w}}.$$

Differentiating once more and evaluating at $(0, w_0)$ gives

$$\begin{aligned}\partial_k B &= -(1 - \Phi(d)), & \partial_w B &= \frac{\varphi(d)}{2\sqrt{w_0}}, \\ \partial_{kk}B &= -(1 - \Phi(d)) + \frac{\varphi(d)}{\sqrt{w_0}}, & \partial_{kw}B &= \frac{\varphi(d)}{4\sqrt{w_0}}, \\ \partial_{ww}B &= -\frac{\varphi(d)}{4w_0^{3/2}} - \frac{\varphi(d)}{16\sqrt{w_0}}.\end{aligned}\tag{60}$$

By (30)–(32),

$$c'(0) = -(1 - u_*), \quad c''(0) = f_* - (1 - u_*).$$

One derivative of the defining identity gives $\partial_k B + \partial_w B w'(0) = c'(0)$, hence

$$w'(0) = \frac{-(1 - u_*) + (1 - \Phi(d))}{\varphi(d)/(2\sqrt{w_0})} = \frac{2\sqrt{w_0}}{\varphi(d)}\Delta.$$

Two derivatives give

$$\partial_{kk}B + 2\partial_{kw}B w' + \partial_{ww}B (w')^2 + \partial_w B w'' = c''.$$

Solving at the money and using (60),

$$\begin{aligned} w''(0) &= \frac{2\sqrt{w_0}}{\varphi(d)} \left[f_* + \Delta - \frac{\varphi(d)}{\sqrt{w_0}} - \frac{\varphi(d)}{2\sqrt{w_0}} w'(0) \right. \\ &\quad \left. + \left(\frac{\varphi(d)}{4w_0^{3/2}} + \frac{\varphi(d)}{16\sqrt{w_0}} \right) w'(0)^2 \right] \\ &= \frac{2\sqrt{w_0}}{\varphi(d)} f_* - 2 + \left(\frac{1}{2w_0} + \frac{1}{8} \right) w'(0)^2. \end{aligned}$$

The terms linear in Δ cancel because $w'(0) = 2\sqrt{w_0}\Delta/\varphi(d)$.

Finally, $\sigma = \sqrt{w/\tau}$ gives

$$\sigma' = \frac{w'}{2\sqrt{\tau w}}, \quad \sigma'' = \frac{w''}{2\sqrt{\tau w}} - \frac{(w')^2}{4\sqrt{\tau} w^{3/2}}.$$

Substitution yields (39)–(40).

A.3 Proof of universality and the closure dichotomy

Let g^* satisfy proposition 3.3, write $\lambda_+^* = e^{g^*(1)} < 1$, and fix $\bar{\lambda} \in (\lambda_+^*, 1)$.

Polynomial approximation. By Weierstrass there are polynomials p_N with $\varepsilon_N = \|p_N - g^*\|_\infty \rightarrow 0$. Every polynomial of degree at most N is represented by the endpoint chord and $P_2, \dots, P_N$. For all large N ,

$$p_N(1) \leq g^*(1) + \varepsilon_N < \log \bar{\lambda} < 0,$$

so the approximating slice is admissible.

Transport convergence and a common envelope. Let $\bar{x}_N$ be the corresponding unshifted transport. On every finite z -interval,

$$|\bar{x}_N(z) - \bar{x}^*(z)| \leq (e^{\varepsilon_N} - 1)e^{\|g^*\|_\infty} |z|,$$

and therefore $\bar{x}_N \rightarrow \bar{x}^*$ locally uniformly. Continuity of g^* at the right endpoint gives z_0 such that, for all large N , $e^{p_N(\Lambda(z))} \leq \bar{\lambda}$ when $z \geq z_0$. Hence a constant C_1 , independent of N , satisfies

$$e^{\bar{x}_N(z)} \leq C_1 e^{\bar{\lambda}z} \quad (z \geq 0), \quad e^{\bar{x}_N(z)} \leq C_1 \quad (z \leq 0). \quad (61)$$

Normalizers, shifts, and quantiles. The functions $e^{\bar{x}_N} \rho$ converge pointwise and, by (61), are dominated by integrable multiples of $e^{(\bar{\lambda}-1)z}$ on the right and e^z on the left. Dominated convergence gives $I_N \rightarrow I^*$, so $m_N = -\log I_N \rightarrow m^*$. Thus $x_N \rightarrow x^*$ locally uniformly and $Q_N \rightarrow Q^*$ uniformly on compact subsets of $(0, 1)$.

Uniform convergence of calls. Couple all laws with one uniform rank and let $Y_N = e^{Q_N(U)}$, $Y^* = e^{Q^*(U)}$. Since positive-part payoffs are one-Lipschitz,

$$\sup_k |c_N(k) - c^*(k)| \leq \mathbb{E}|Y_N - Y^*|.$$

The integrand converges pointwise. The envelope above and boundedness of m_N give a constant C_2 such that $e^{Q_N(u)} \leq C_2 \{1 + (1-u)^{-\bar{\lambda}}\}$. This is integrable, so the calls converge uniformly.

Wasserstein convergence. The quantile coupling is optimal on the line:

$$W_p(Y_N, Y^*)^p = \int_0^1 |e^{Q_N(u)} - e^{Q^*(u)}|^p du.$$

For $1 \leq p < 1/\lambda_+^*$ choose $\bar{\lambda} \in (\lambda_+^*, 1/p)$. The integrand is dominated by a multiple of $1 + (1-u)^{-p\bar{\lambda}}$, proving the claim by dominated convergence.

Finite-order exclusions. At finite order, x'_N is positive and continuous. The quantile is a strictly increasing homeomorphism from $(0, 1)$ onto $\mathbb{R}$; it can neither jump, remain flat, nor terminate. This excludes atoms, density gaps, bounded support, and mass at zero. A Gaussian-or-faster tail would require $g(u) \rightarrow -\infty$ at an endpoint, which no polynomial can do.

Uniform closure away from the wall. Suppose admissible g_N converge uniformly to g_∞ . Necessarily $g_\infty(1) \leq 0$. If $g_\infty(1) < 0$, the preceding dominated-convergence argument applies with g_∞ as target. The limit retains a positive continuous density and exponential tails.

The boundary branch. If $g_\infty(1) = 0$, weak convergence may fail. For $g_N \equiv \log(1 - 1/N)$, the solvable slice gives

$$I_N = \frac{\pi(1 - 1/N)}{\sin(\pi/N)} \rightarrow \infty, \quad m_N = -\log I_N \rightarrow -\infty.$$

Every fixed-rank quantile tends to $-\infty$, so mass escapes.

Assume instead that the laws converge weakly. The shifts are bounded above because every normalizer has a positive central contribution. They are bounded below because $m_N \rightarrow -\infty$ along a subsequence would contradict tightness. Every convergent subsequence of shifts then yields

$$Q_\infty(u) = m_\infty + \int_0^{\text{logit } u} e^{g_\infty(\Lambda(t))} dt.$$

Weak convergence identifies this quantile uniquely, so the full shift sequence converges. The limiting speed is continuous and bounded away from zero; hence the limit has full support and a strictly positive continuous density, and all geometric exclusions remain.

The right endpoint speed is one. For each $\delta > 0$ it is eventually at least $1 - \delta$, so the argument of proposition 4.2 gives $\mathbb{E}[e^{rX}] = \infty$ for every $r > 1$. Fatou also gives $\mathbb{E}[e^X] \leq \liminf_N \mathbb{E}[e^{X_N}] = 1$, with equality not guaranteed. This is the boundary law in proposition 3.3.

B Numerical implementation

This appendix records the load-bearing parts of a reference implementation. The pseudocode follows the notation of the chapter; tail-sensitivity terms and error handling are described after the listings.

B.1 Slice build

Listing 1: Build the transport and ledger on one fixed grid.

```
def build(theta, grid):
    L, R, a = theta.L, theta.R, theta.a
    g = (1 - grid.u)*L + grid.u*R + a @ grid.Pmat
    if not finite(g) or max(g) > 700:
        raise InteriorOverflow

    dx = exp(g) # x'(z)
```

```

xbar = cumulative_simpson(dx, anchor=grid.mid)
if max(xbar) > 700:
    raise InteriorOverflow

lam_m, lam_p = exp(g[0]), exp(g[-1])
if lam_p >= 1 - 1e-6:
    raise InadmissibleTail

mass = exp(xbar) * grid.rho
I = integrate(mass) \
    + exp(xbar[-1] - grid.zmax)/(1 - lam_p) \
    + exp(xbar[0] - grid.zmax)/(1 + lam_m)
m = -log(I)
x = m + xbar

dG = -exp(x) * grid.rho # G'(z)
G = reverse_integrate(-dG) \
    + exp(x[-1] - grid.zmax)/(1 - lam_p)
return Slice(x=x, dx=dx, G=G, dG=dG,
             m=m, lam=(lam_m, lam_p), theta=theta)

```

The cached density must be computed as $\expit(z)\expit(-z)$. Forming $u*(1-u)$ loses the right tail when u rounds to one. Both x and G carry their exact nodal derivatives, so a cubic Hermite representation needs no global spline solve.

The median is the natural anchor for the cumulative transport. It fixes $\bar{x}(0) = 0$ and lets a single forward cumulative sum build the positive half-line while the same stencil, reversed, builds the negative half-line. The additive freedom is then carried entirely by m . Simpson quadrature is fourth order on the uniform grid; the Hermite interpolant has the same local order because both values and exact nodal derivatives are available.

The grid does not truncate the probability law. From (16), on the right

$$e^{\bar{x}(z)}\rho(z) = e^{b_+ + (\lambda_+ - 1)z} \{1 + O(e^{-z})\}.$$

Integrating its leading term from $Z_{\max}$ gives the first correction in (53); the relative error is $O(e^{-Z_{\max}})$. The left correction follows from $e^{\bar{x}(z)}\rho(z) \sim e^{b_- + (\lambda_- + 1)z}$. The factor $1 - \lambda_+$ in the right denominator is also a numerical warning: tail mass becomes increasingly sensitive as the finite-forward wall is approached.

B.2 Call evaluation

Listing 2: Evaluate a normalized call.

```

def call_price(slc, k, grid):
    if k > slc.x[-1]:
        z = grid.zmax + (k - slc.x[-1]) / slc.lam[1]
        return exp(k - z) * slc.lam[1] / (1 - slc.lam[1])
    if k < slc.x[0]:
        return 1 - exp(k) + put_asymptote_left(slc, k, grid)

    z = hermite_root(slc.x, slc.dx, k)
    share = hermite_eval(slc.G, slc.dG, z)
    cash = exp(k - logaddexp(0, z)) # e^{k(1-Lambda(z))}
    return share - cash

```

The log-space cash expression is essential. The algebraically simpler $\exp(k)*(1-\expit(z))$ becomes zero after rank saturation and can create a spurious upward step in the far call wing.

The two branches beyond the return grid are part of the same law. On the right, (54) continues the exact power tail. On the left, put asymptotics are computed first and call parity is applied. This avoids subtracting two quantities close to one when a deep in-the-money call is evaluated directly.

B.3 Sensitivity pass

Listing 3: All raw-coordinate Jacobian columns on one build.

```
def sensitivities(slc, grid):
    # rows: dg/dL, dg/dR, dg/da_2, ..., dg/da_N
    B = vstack([1-grid.u, grid.u, grid.Pmat])
    dxbar = cumulative_simpson(slc.dx * B,
                              anchor=grid.mid, axis=1)
    weight = exp(slc.x) * grid.rho
    dm = -integrate(weight * dxbar, axis=1)
    dx = dm[:, None] + dxbar

    integrand = weight[None, :] * dx
    dG = reverse_integrate(integrand, axis=1)
    dG_slope = -integrand
    dvs = -2 * integrate(dx * grid.rho, axis=1)
    return dG, dG_slope, dvs

def price_jacobian_row(slc, sens, k):
    z = hermite_root(slc.x, slc.dx, k)
    return hermite_eval(sens.dG, sens.dG_slope, z)
```

The derivatives of the two tail corrections in (53) must also be included. They use the endpoint sensitivities

$$\partial\lambda_-/\partial\theta = \lambda_-(1, 0, 1, 1, \dots), \quad \partial\lambda_+/\partial\theta = \lambda_+(0, 1, 1, -1, \dots).$$

Derivatives in the endpoint or logistic chart follow by multiplication with the corresponding chart Jacobian.

The analytic pass performs one cumulative integral for each basis column, one normalization dot product, and one reverse cumulative integral. Its work is $O(Pn_{\text{grid}})$ and its storage may be either $O(Pn_{\text{grid}})$, when all columns are retained, or $O(n_{\text{grid}})$ per streamed column. Finite differences have the same formal order but repeat normalization, ledger construction, and every strike root $P + 1$ times. Envelope cancellation removes the roots from the analytic derivative and is therefore responsible for most of the measured constant-factor gain.

B.4 Interpolation certificate

Listing 4: Fritsch–Carlson margin for a uniform-grid Hermite interpolant.

```
def monotone_margin(y, d, dz, flat_tol=1e-9):
    sec = diff(y) / dz
    d0, d1 = d[:-1], d[1:]
    active = ~((abs(sec) < flat_tol) &
               (abs(d0) < flat_tol) &
               (abs(d1) < flat_tol))
    margin = minimum.reduce([d0, d1,
                             3*sec-d0, 3*sec-d1])
    return min(margin[active])
```

An increasing interpolant is certified when the returned margin is positive. Apply the same routine to $-G$. Segments excluded by `flat_tol` are recorded as flat to tolerance rather than strictly monotone.

For a cubic Hermite segment, let s be its secant slope and d_0, d_1 its endpoint derivatives. The sufficient Fritsch–Carlson box is

$$0 \leq d_0 \leq 3s, \quad 0 \leq d_1 \leq 3s.$$

The routine returns the minimum distance to the four faces of this box. A positive number is therefore not merely a pass flag but a margin. For smooth functions sampled on a fine grid, $d_i/s = 1 + O(h^2)$, so the certificate should clear the boundary comfortably. A small margin is a useful indication of under-resolution even before monotonicity fails.

Monotonicity of x and G is not by itself a convexity certificate for the interpolated call. The separate belly test uses the exact factorization (5): analytic w, w', w'' are evaluated at 801 points across the retained strike span and g_D must exceed -10^{-4} . Since a flat smile has $g_D = 1$, the tolerance permits a negative density component no larger than 10^{-4} of the local Black density factor. The threshold is therefore dimensionless and interpretable.

B.5 Floating-point failure catalogue

Four failures deserve explicit tests.

1. *Rank saturation.* For z near 37, double precision rounds $\text{expit } z$ to one. Any formula that subsequently forms $1 - u$ loses the right-tail probability. Use $\text{expit}(-z)$ directly and keep the cash leg in log space.
2. *Interior overflow.* Admissible endpoints do not bound a polynomial between them. Large Legendre coefficients can make g exceed the exponent range in the body while $g(0)$ and $g(1)$ remain moderate. The build must inspect the full grid and reject the trial before exponentiation.
3. *Near-wall cancellation.* Normalization and right-tail prices contain $1/(1 - \lambda_+)$. The logistic chart slows parameter steps near the wall, while the hard guard leaves a finite numerical margin.
4. *Lee cancellation.* In the textbook expression for $\Psi(r)$, $\sqrt{r^2 + r} - r$ subtracts nearly equal quantities for large r . Use the rationalized form (55), with the limiting value zero handled explicitly as $r \rightarrow \infty$.

Cached, parameter-independent grid arrays should be read-only. An accidental in-place modification of ranks, logistic weights, or basis values otherwise corrupts every later slice without necessarily producing an exception.

B.6 Validation stack

The solvable slice tests the most code with the least ambiguity. Across $s \in [0.05, 0.95]$, compare the computed transport with $m + sz$, the shift with $-\log(\pi s / \sin \pi s)$, and the martingale integral with one. The observed worst errors are 1.1×10^{-11} , 3.0×10^{-9} , and 3.3×10^{-16} . The shift error grows toward the wall because the normalizing integrand decays at rate $1 - s$.

The randomized battery serves a different purpose. It samples several basis orders, ordinary slices, near-wall slices, and slices with large interior modes. Strikes are placed between grid nodes, and convexity is checked in normalized strike rather than log-strike. A four-times-finer independent build provides a reference price. Together with the interpolation and belly certificates, this separates algebraic validity, interpolation validity, and grid accuracy.

B.7 Reference defaults and timing

The exact timings depend on hardware and caches. The portable facts are the $O(Pn_{\text{grid}})$ analytic complexity and the removal of repeated builds. The reported medians and interquartile ranges come from a single run that alternates analytic and finite-difference solves after warm-up. Alternation reduces drift from cache state and background load; medians avoid a best-of- n claim. A timing table should always travel with its grid size, quote count, order, solver tolerances, and machine context.

Quantity	Default	Quantity	Default
Basis order N	16 (cap 24)	Order guard	(50)
Grid	$[-40, 40]$	Nodes	8001 (fit: 2001)
Wall guard	$\lambda_+ \leq 1 - 10^{-6}$	Exponent budget	700
Ridge (α, ν)	$(10^{-6}, 1), n \geq 4$	Vega floor η	10^{-4}
Tail barrier	center 0.90, slope 50	Haircut h	0.005 vol
Solver	trust-region LSQ	Tolerances	10^{-10}
Chart	logistic	Calendar nodes	$\max(49, 4N + 1)$
Belly check	801 points	Cold start	$\sqrt{3w_0}/\pi$

Table 3: Reference implementation defaults.

Configuration	Order	Median	IQR	vs. finite difference
Analytic Jacobian	6	15.9 ms	0.5 ms	$1.41 \times$
Finite difference	6	22.5 ms	2.7 ms	—
Analytic Jacobian	12	26.0 ms	2.1 ms	$1.68 \times$
Finite difference	12	43.8 ms	3.2 ms	—
Analytic Jacobian	16	28.9 ms	0.5 ms	$1.76 \times$
Finite difference	16	50.9 ms	3.9 ms	—

Table 4: Single-slice calibration times from one interleaved run. The last column is the median finite-difference time divided by the median analytic time for the same configuration.

For a new implementation, the shortest reliable porting sequence is:

1. reproduce the constant-speed normalizer and transport;
2. implement the stable cash leg and tail corrections;
3. keep (x, x', G, G') as one immutable slice object;
4. add the envelope sensitivity pass;
5. run monotonicity, Durrleman, and randomized sub-grid checks on every release.

C Data and reproducibility

Every empirical number and figure is generated from one frozen JSON snapshot, captured after the 2026-08-03 US session from a delayed consolidated US listed-options feed. The file contains SPY and NVDA chains for eight expiries each, with spots 757.67 and 206.80, together with forwards, discount factors, prepared quotes, fitted parameters, and display curves. Maturities run from 2 calendar days to 1.3726 years.

The snapshot is self-contained. For each node it records the normalized chain, forward and discount provenance, retained quote set, fitted raw parameters, and diagnostic curves. Reproduction therefore requires neither credentials nor a market-data connection. The capture timestamp, reference date, and session are part of the manifest, so a later book cannot silently replace the one used for the chapter.

The manifest records the common fitting recipe: order 16 under (50), logistic chart, haircut $h = 0.005$, equal quote weights, ridge $\alpha = 10^{-6}$ with $\nu = 1$, and calendar rows active. All 16 nodes were fitted; none was tuned individually. The shortest NVDA node has the largest residual, consistent with its sparse and steep quote strip. SPY bands collapse to mids under the haircut rule; NVDA retains interval targets. These states are stored per quote rather than reconstructed from rounded display values.

The figure generator reads only this snapshot and fixed synthetic specifications. Real nodes are rebuilt from the frozen coefficients rather than refitted, and the rebuilt curves agree with the

stored curves to 0.000 vol bp in the worst case. The script emits both the 14 figure PDFs and the LaTeX macro file supplying every empirical value in the chapter. Randomized audits use fixed seeds. Timing values follow a fixed interleaved protocol but vary with the machine.

The synthetic double-hump target is a fixed two-component lognormal mixture, martingale-normalized and sampled on a realistic strike ladder. The constant-speed benchmark requires no data and is evaluated directly from (18). Regeneration consists of running the figure generator and then compiling the present manuscript. Refetching the market feed would create a different dataset and is not part of reproduction.

One family-level diagnostic is retained with the data. The two shortest SPY expiries cross in implied-volatility space only far outside their common quote span, at call prices near numerical zero. Their direct normalized call prices remain ordered to 2.2×10^{-15} on the audited grid. Keeping both measurements makes the support-confinement example of section 8 reproducible without interpreting the picture by eye.

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